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Department of CSE

B.Tech. CSE Trimester -5

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Section: 3

Cloud Infrastructure and Services
25CS2206E

PRACTICAL RECORD

2026

Practical -2

AIM: To enable Multi-Factor Authentication (MFA) for secure access to the AWS account, create a first Linux instance in Amazon EC2 and connect to it, and create an S3 bucket and upload five objects with different file extensions.

Description:

Multi-Factor Authentication (MFA) adds a second verification step to the normal user name and password. After MFA is enabled, the account can be accessed only when the password and the second factor are both provided. AWS supports passkeys and security keys as MFA devices, which use the fingerprint, face unlock or screen lock of the device and give strong protection against phishing and password theft. Since the root user has full access to the account, enabling MFA for the root user is the first security step.

Amazon Elastic Compute Cloud (EC2) provides resizable computing capacity in the cloud. A Linux instance is created from an Amazon Machine Image (AMI), which contains the operating system and the software required to start the instance. The Amazon Linux 2023 AMI is free tier eligible and the t3.micro instance type can be used without charges within the free tier limits. A key pair and a security group are selected while launching, and the instance can be connected from the browser using EC2 Instance Connect. The instance is terminated after the work is completed so that no charges are incurred.

Amazon Simple Storage Service (S3) is an object storage service in which data is stored as objects inside buckets. Every bucket name must be unique and is created in a selected AWS Region. Block Public Access keeps the bucket private, and objects are added to the bucket using the Upload option. In this practical, one bucket was created and five objects with different file extensions were uploaded to it, and a second bucket was created to understand unique bucket naming in the global namespace.

Steps:

a) Enabling Multi-Factor Authentication

1. Sign in to the AWS Management Console with the root account and open the IAM service.
2. In My security credentials, scroll to the Multi-factor authentication (MFA) section and click Assign MFA device.
3. Enter a device name, select the Passkey or security key option and continue.
4. Complete the browser prompt and save the passkey in Google Password Manager.
5. Confirm that the MFA table lists the new passkey device with the current date.

b) Creating First Linux Instance

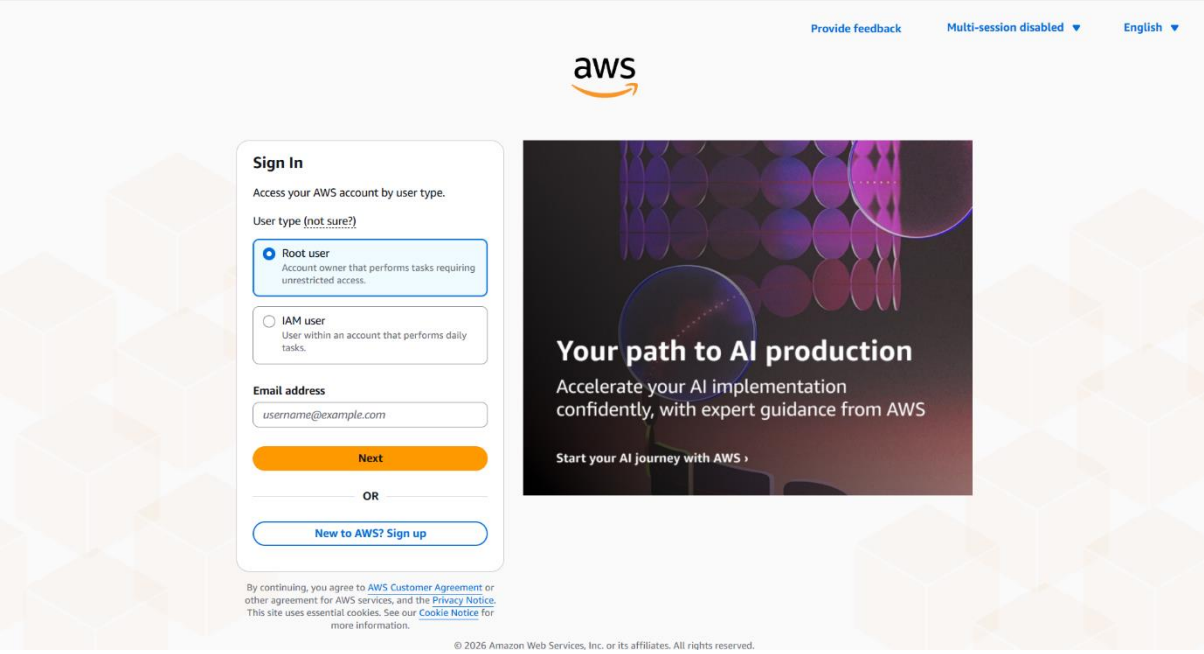
6. Type EC2 in the console search bar and open the EC2 service.
7. On the EC2 Dashboard, click Launch instance.
8. Enter the instance name as PRUDHVI.
9. In Application and OS Images, select the Amazon Linux 2023 AMI (64-bit x86, free tier eligible).
10. Keep the instance type as t3.micro.
11. Select or create a key pair for logging in to the instance.
12. Keep the default VPC and create a new security group that allows SSH access.
13. Leave the storage at 8 GiB gp3 and the advanced details at their default values.
14. Review the summary and click Launch instance.
15. Note the success message and the instance ID i-02a5f5fe0ebec9742.
16. Wait until the instance state is Running and the status check shows 3/3 checks passed.
17. Select the instance, click Connect and choose EC2 Instance Connect with the public IPv4 address.
18. In the browser terminal, run ls, mkdir test, ls, vi file.txt, cat file.txt, history and grep to practise basic Linux commands.
19. Terminate the instance after the commands are verified, so that no charges are incurred.

c) S3 Bucket Creation and Upload Object

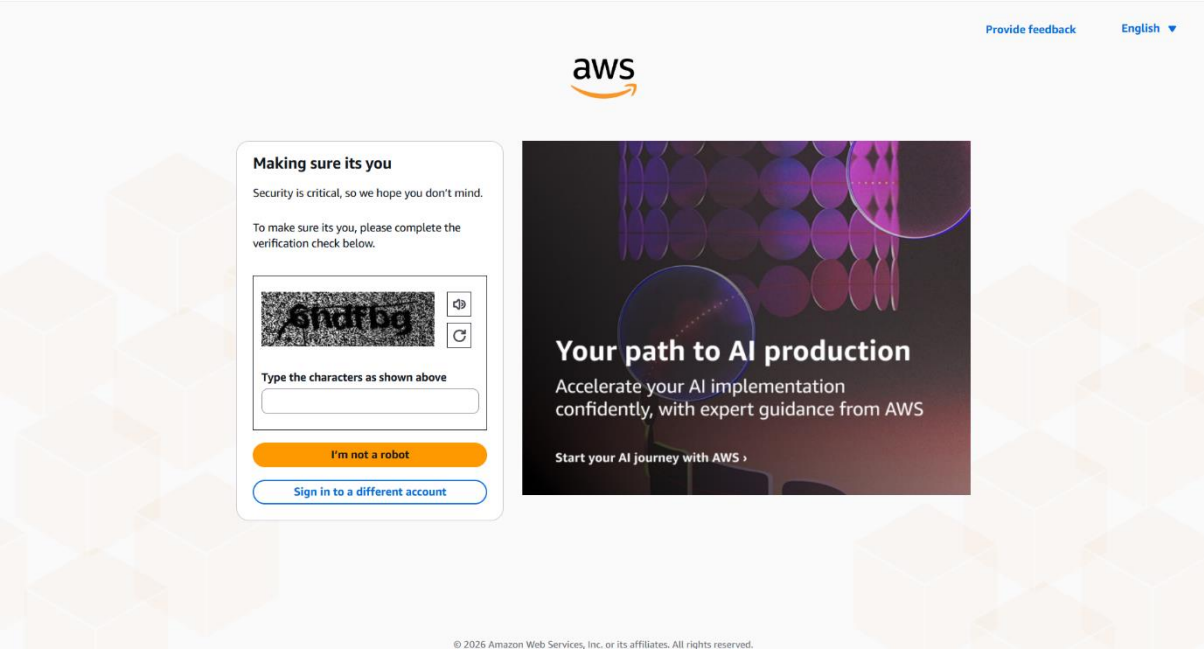
20. Type S3 in the console search bar and open Amazon S3.
21. Click Create bucket and select the Account Regional namespace.
22. Enter the bucket name shiva-s3-bucket-741973715162-eu-north-1-an and select the Europe (Stockholm) region.
23. Keep ACLs disabled and keep Block all public access enabled for the first bucket.
24. Leave versioning disabled and add no tags, then click Create bucket.
25. Open the bucket, click Upload and add five files with different extensions(.pdf, .txt, .jpg, .csv and .docx).
26. Keep the default permissions and Standard storage class, then click Upload.
27. Verify the uploaded objects in the Objects tab of the bucket.
28. Create a second bucket shiva-s3-bucket1-741973715162-eu-north-1-an using the Global namespace and review its public access settings.
29. Open the Buckets page and confirm that both buckets are listed.

Result: Thus, Multi-Factor Authentication was enabled for the AWS account using a passkey, a first Linux instance was launched, connected and terminated successfully, and an S3 bucket was created with five objects of different file extensions. A second bucket was also created to verify unique bucket naming.

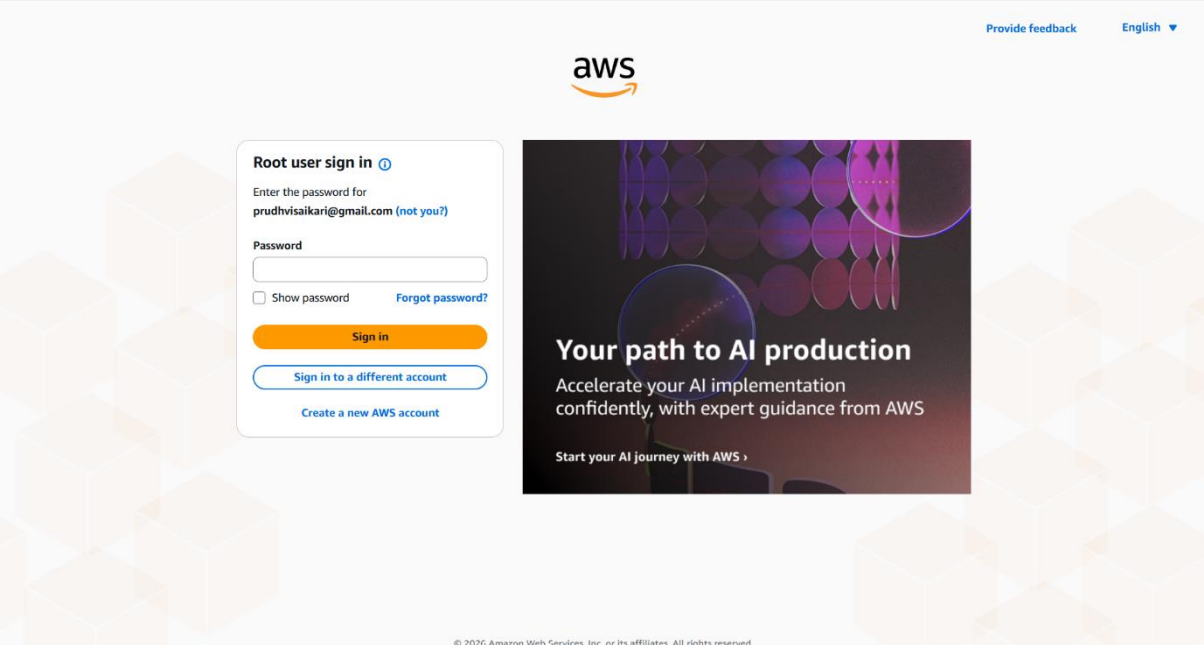
Output:



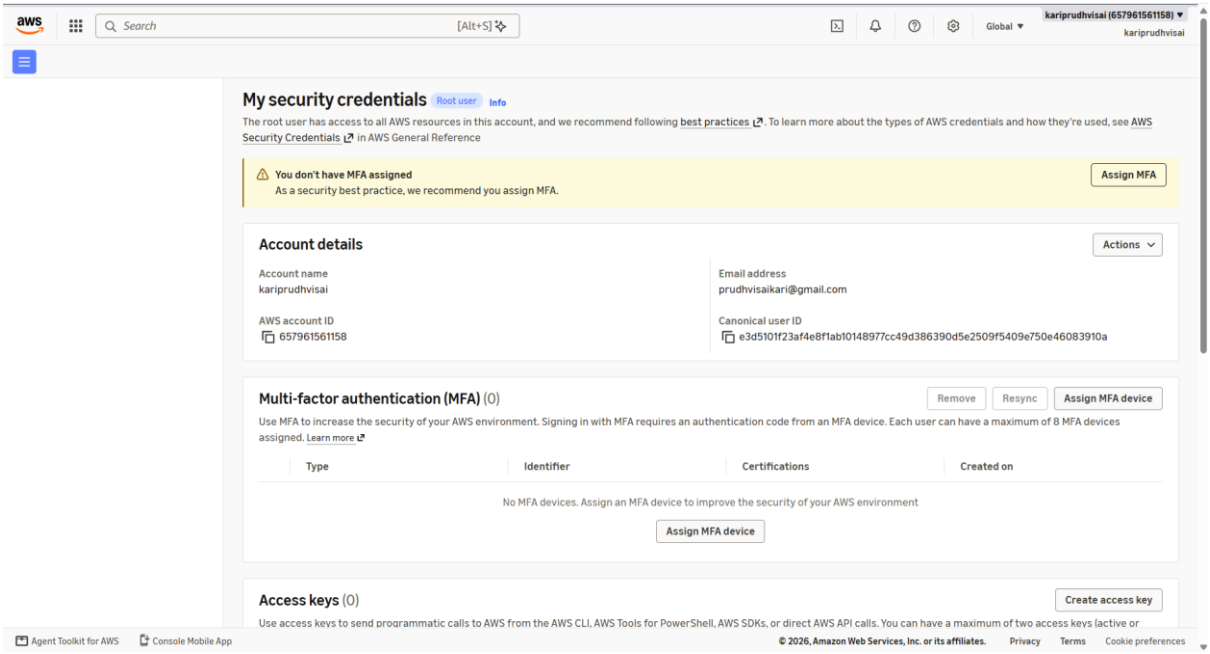
The AWS Sign-In page is displayed, with the "Root user" option selected to access the account.



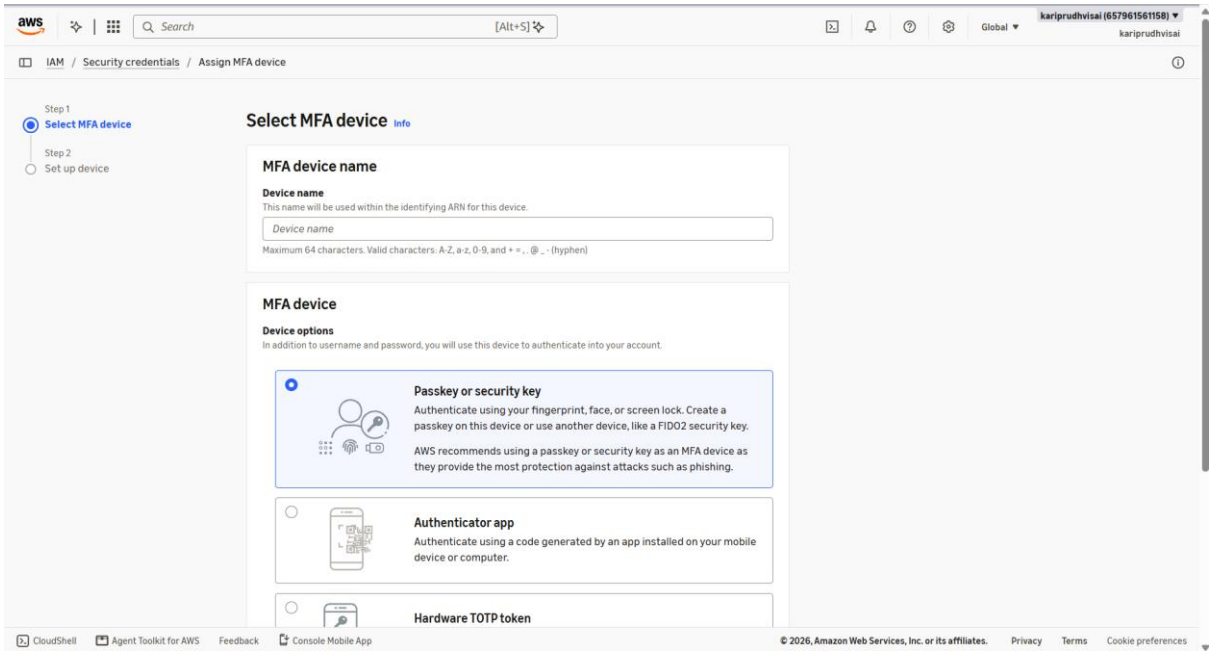
A security verification step requires the user to solve a CAPTCHA before proceeding



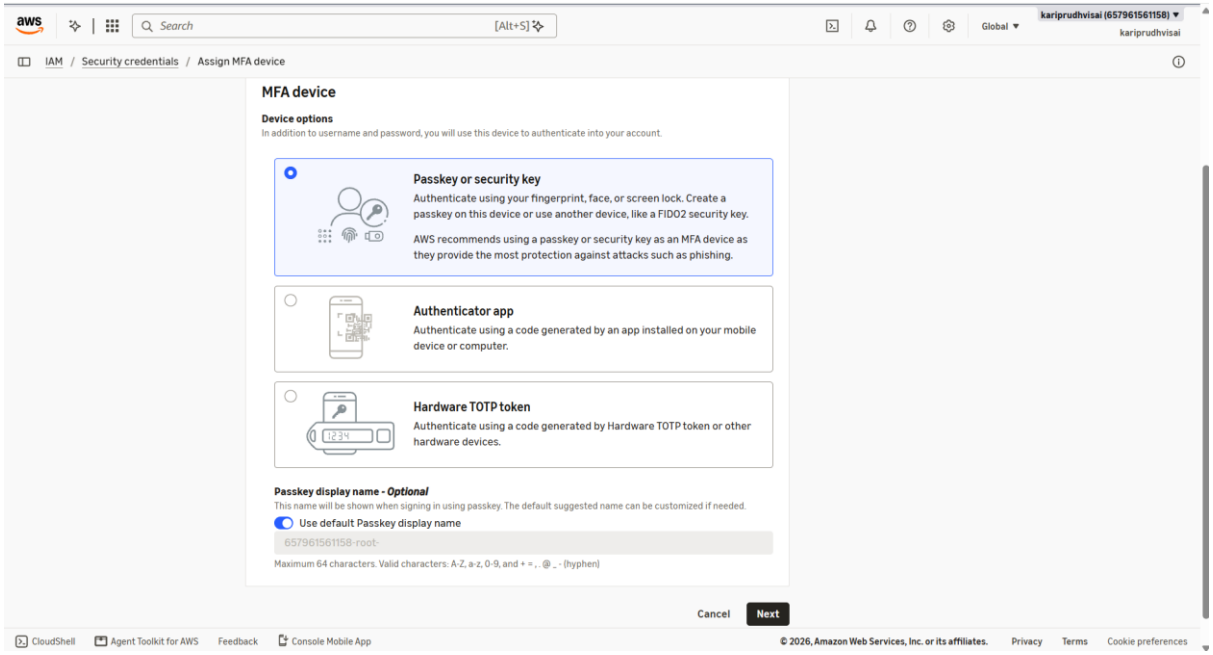
The Root user sign-in page prompts for the password associated with the email address "prudhvisaikari@gmail.com".



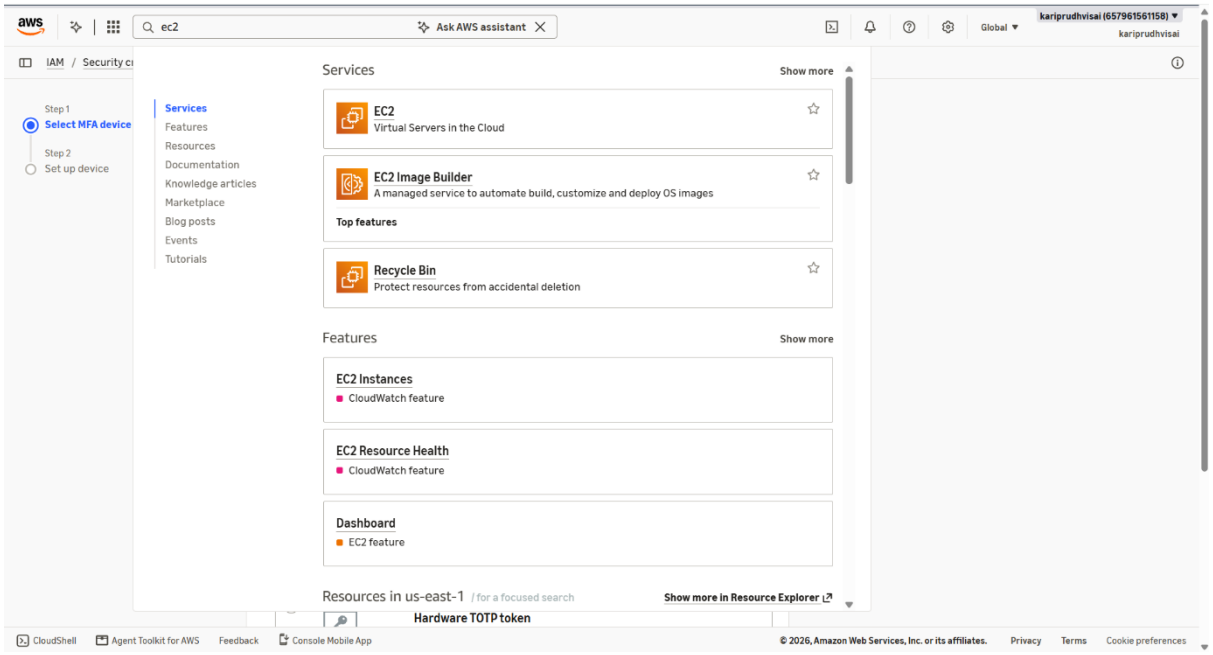
The "My security credentials" dashboard shows the account details for "kariprudhvisai" and displays a prominent warning banner recommending the assignment of Multi-factor authentication (MFA).



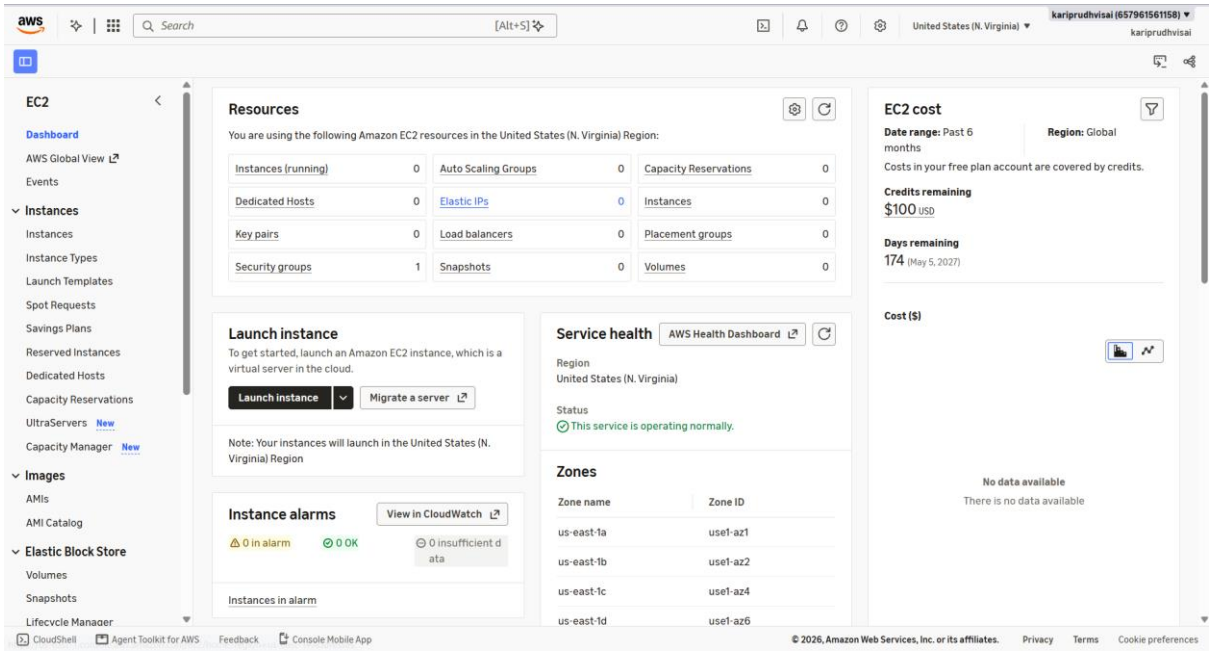
The first step of the "Assign MFA device" process is shown, with the "Passkey or security key" option selected under MFA device options.



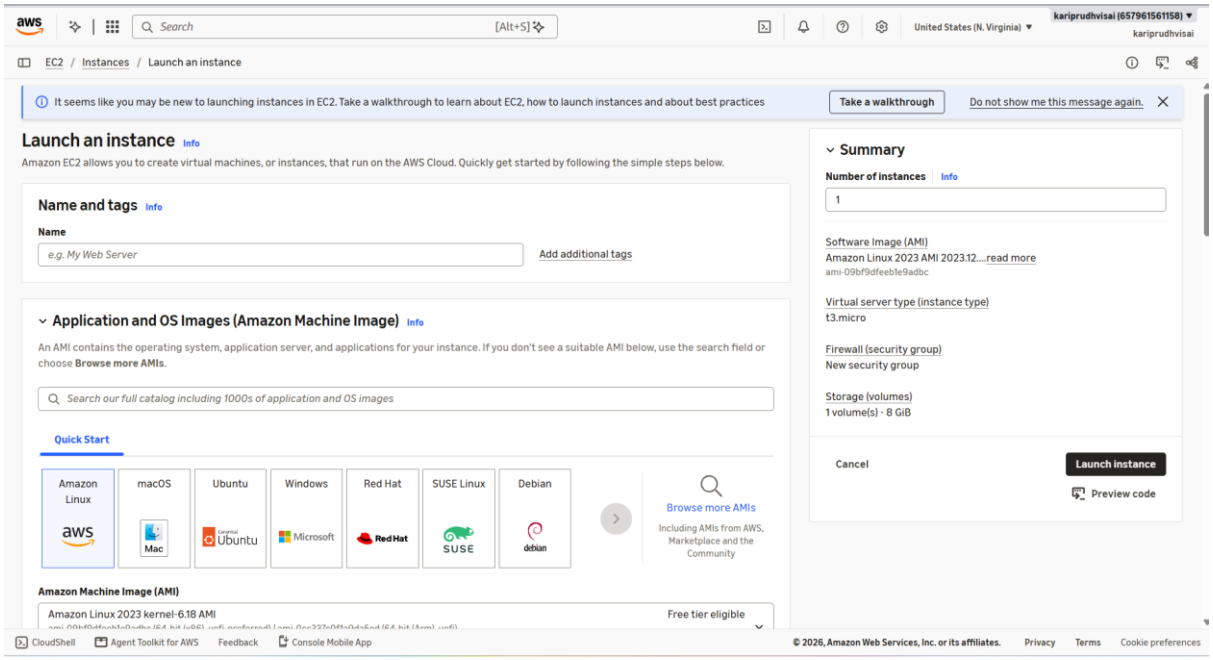
A continuation of the MFA device setup page shows the "Use default Passkey display name" option toggled on, auto-populating a display name using the AWS account ID.



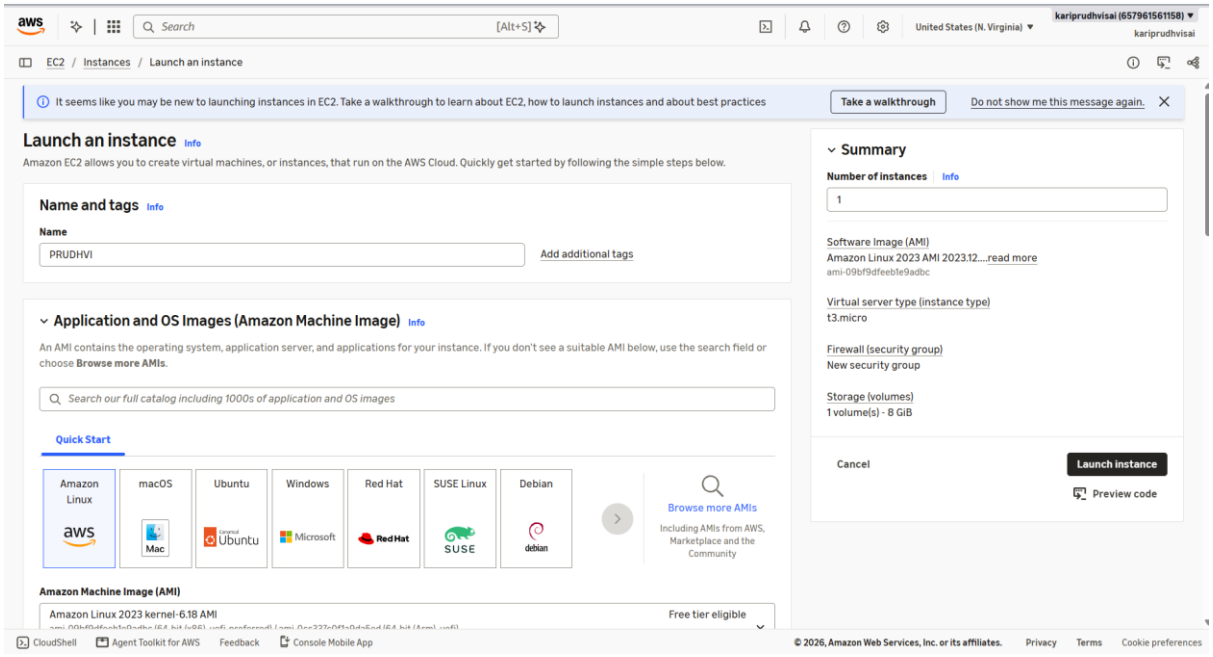
The AWS unified search bar is used to search for "ec2", displaying the "EC2 (Virtual Servers in the Cloud)" service as the top result.



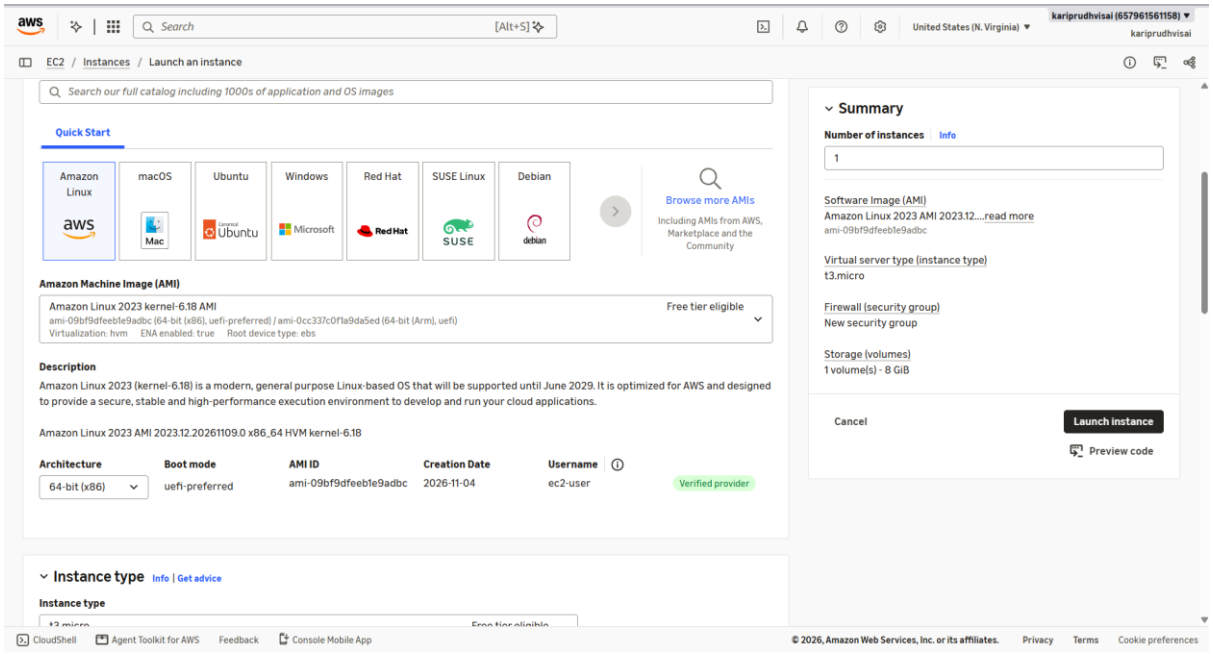
The EC2 Dashboard for the N. Virginia region indicates that there are currently 0 instances running and 1 security group configured.



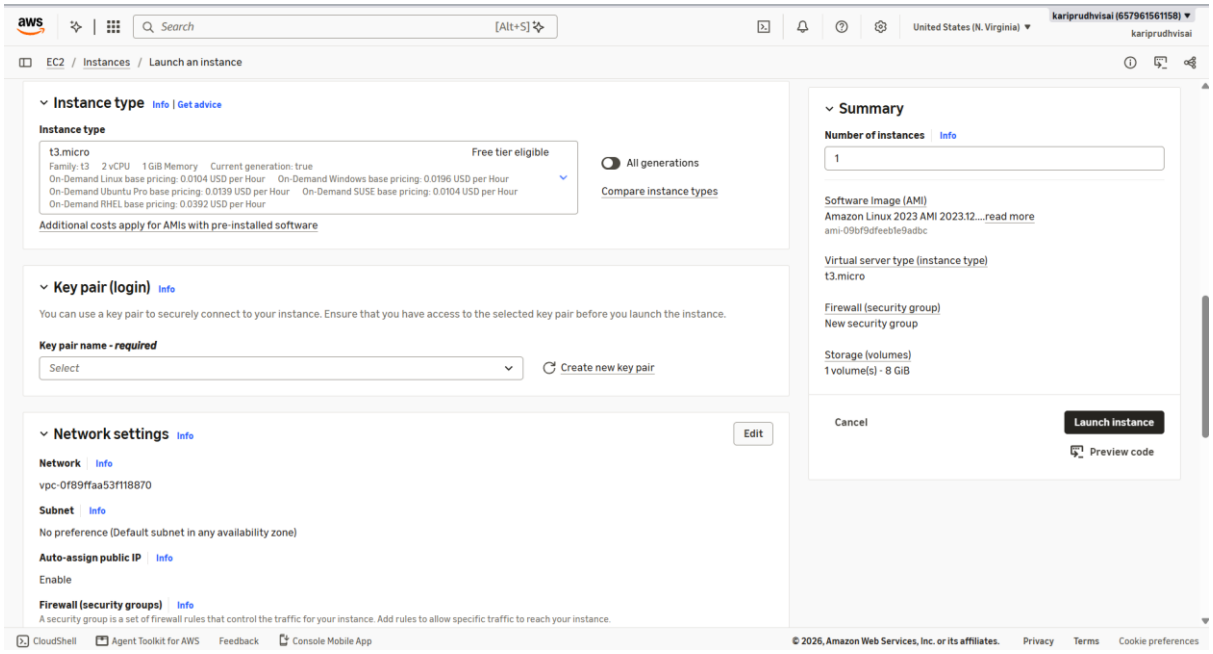
The "Launch an instance" configuration page is opened, showing an empty "Name" field and the default selection of the Amazon Linux machine image.



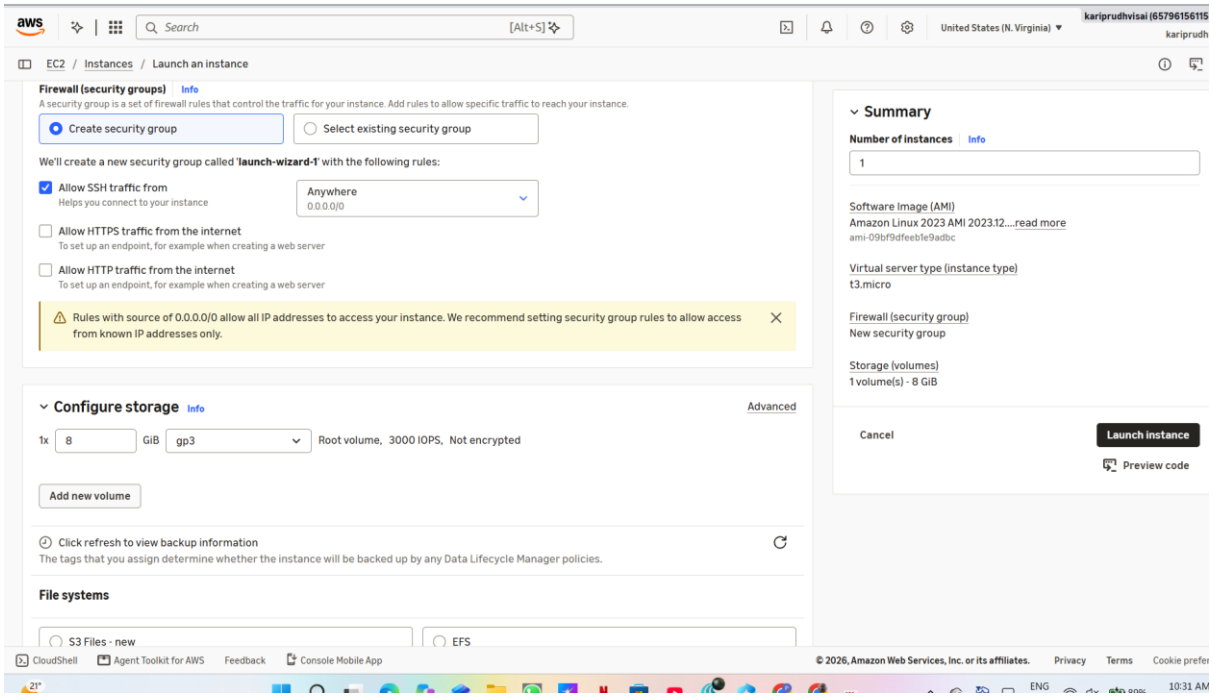
On the same "Launch an instance" page, the "Name" field has been populated with the text "PRUDHVI"



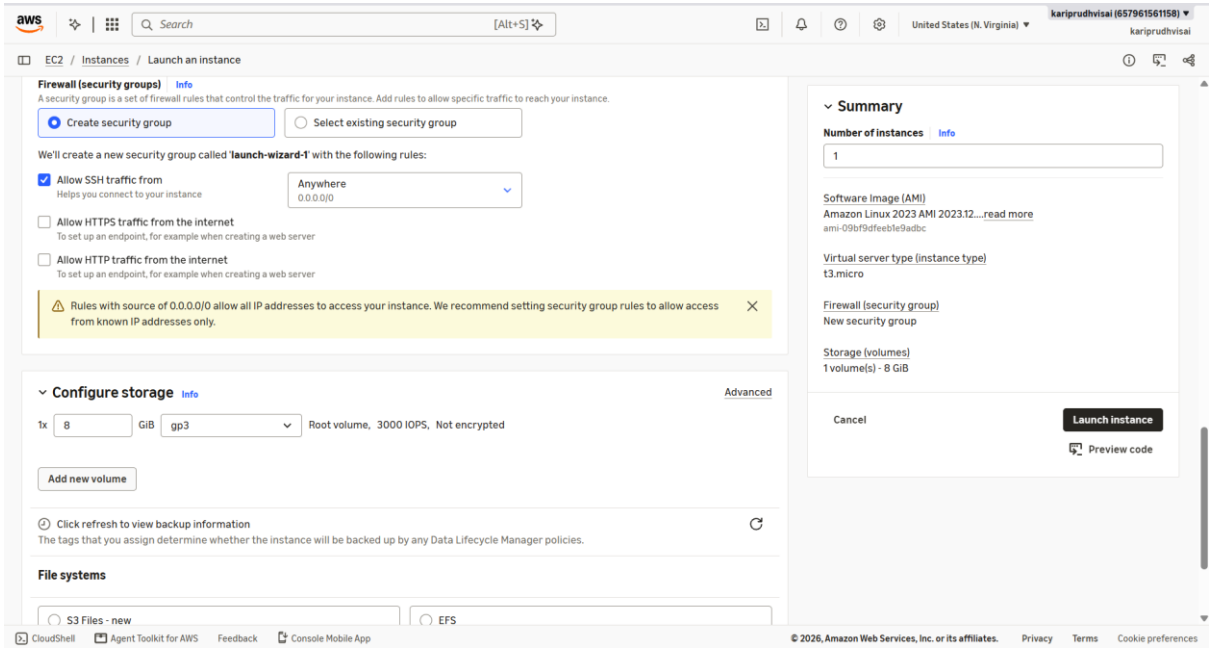
The "Launch an instance" page displays the "Application and OS Images (Amazon Machine Image)" section, where the "Amazon Linux 2023 AMI" is selected. It shows details like the 64-bit (x86) architecture and the AMI ID.



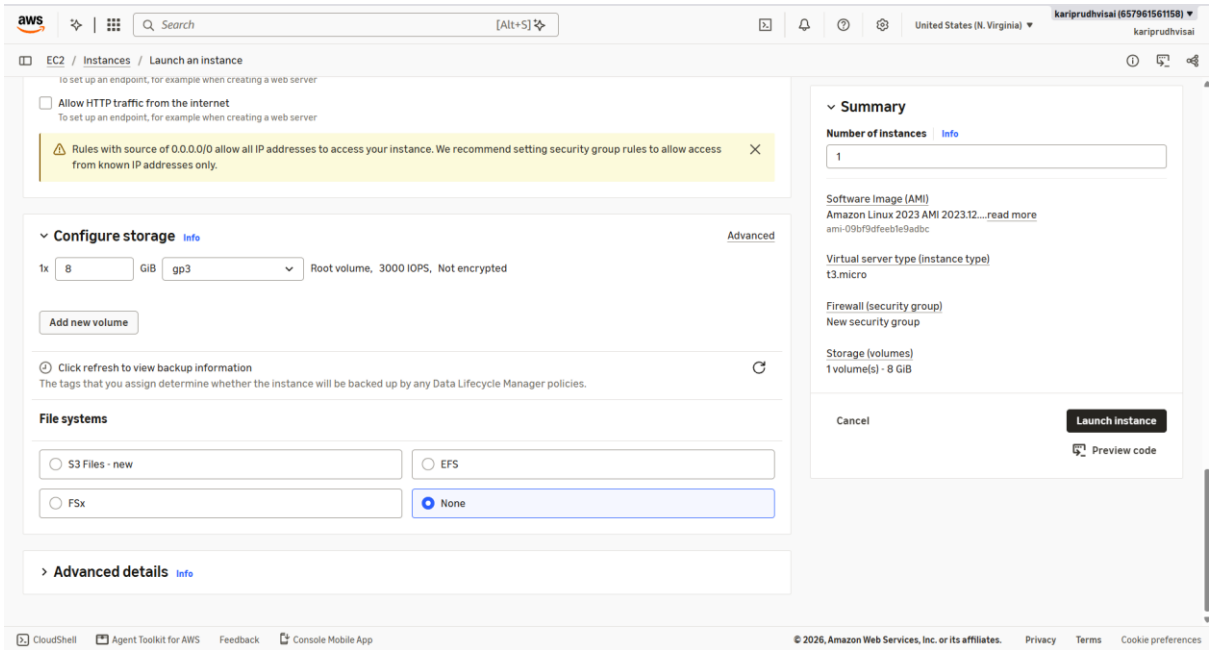
Scrolling down, the "Instance type" is set to "t3.micro", which is marked as "Free tier eligible". The "Network settings" section below it indicates the default VPC and subnet are selected, and Auto-assign public IP is enabled.



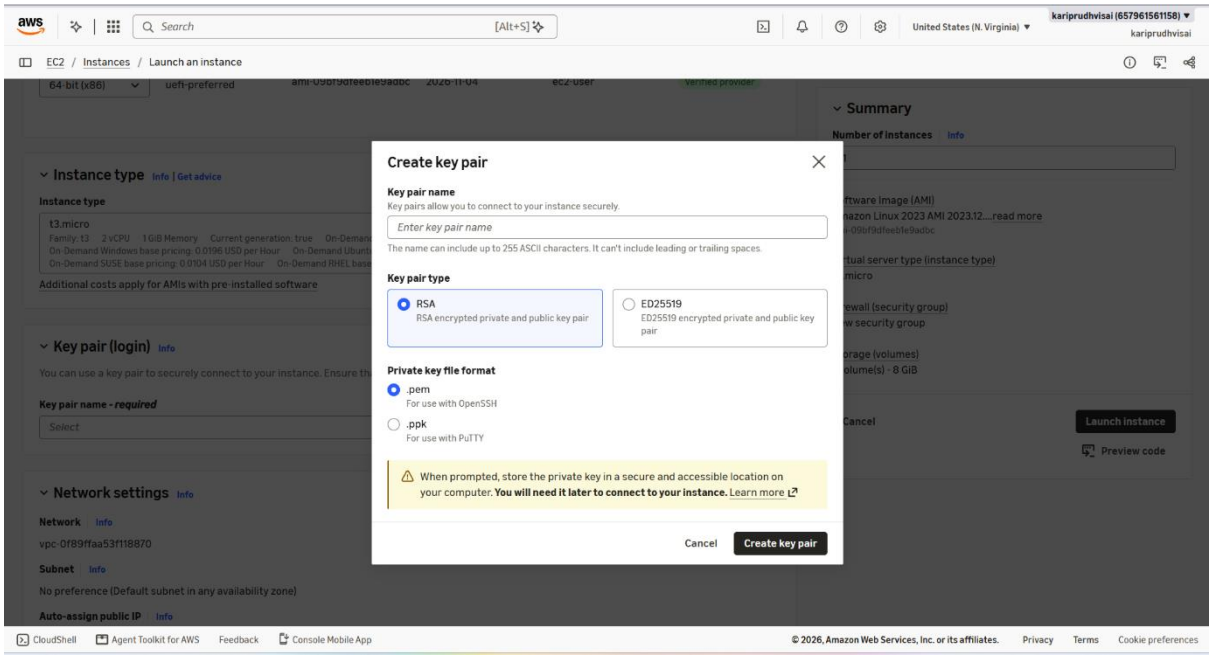
The "Network settings" section is expanded, showing the "Firewall (security groups)" configuration. "Create security group" is selected, and a rule to "Allow SSH traffic from Anywhere" is checked, triggering a warning about allowing all IP addresses.



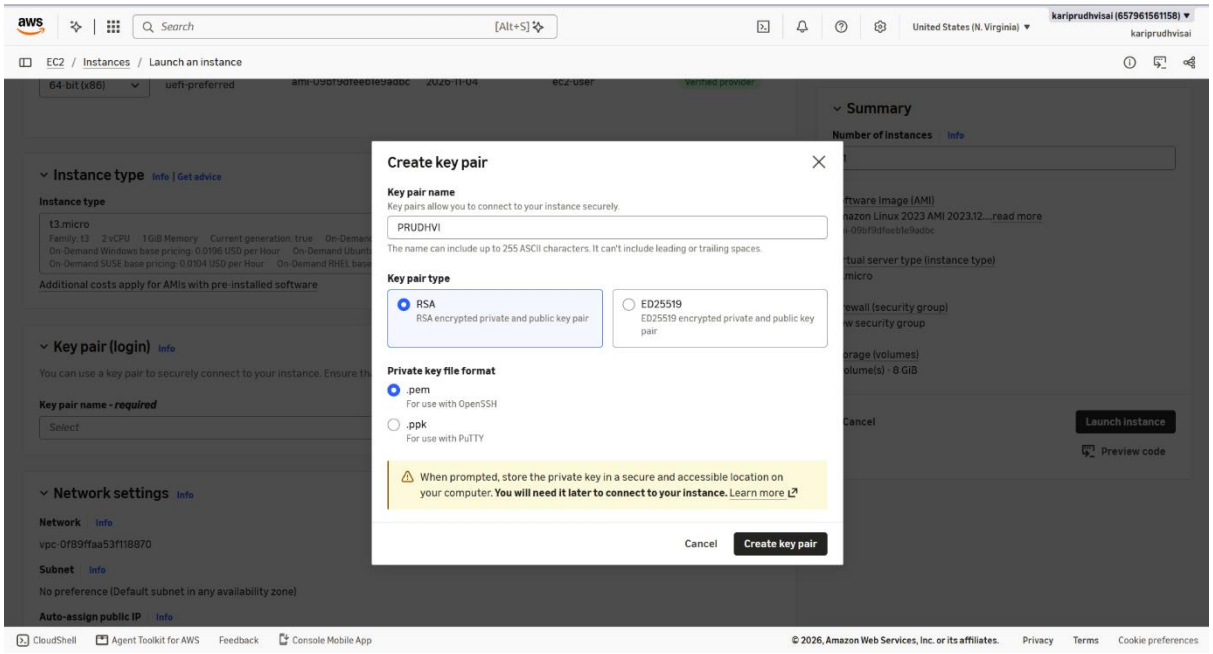
This image is identical to the previous one, showing the "Firewall (security groups)" configuration with the SSH rule enabled from "Anywhere" (0.0.0.0/0).



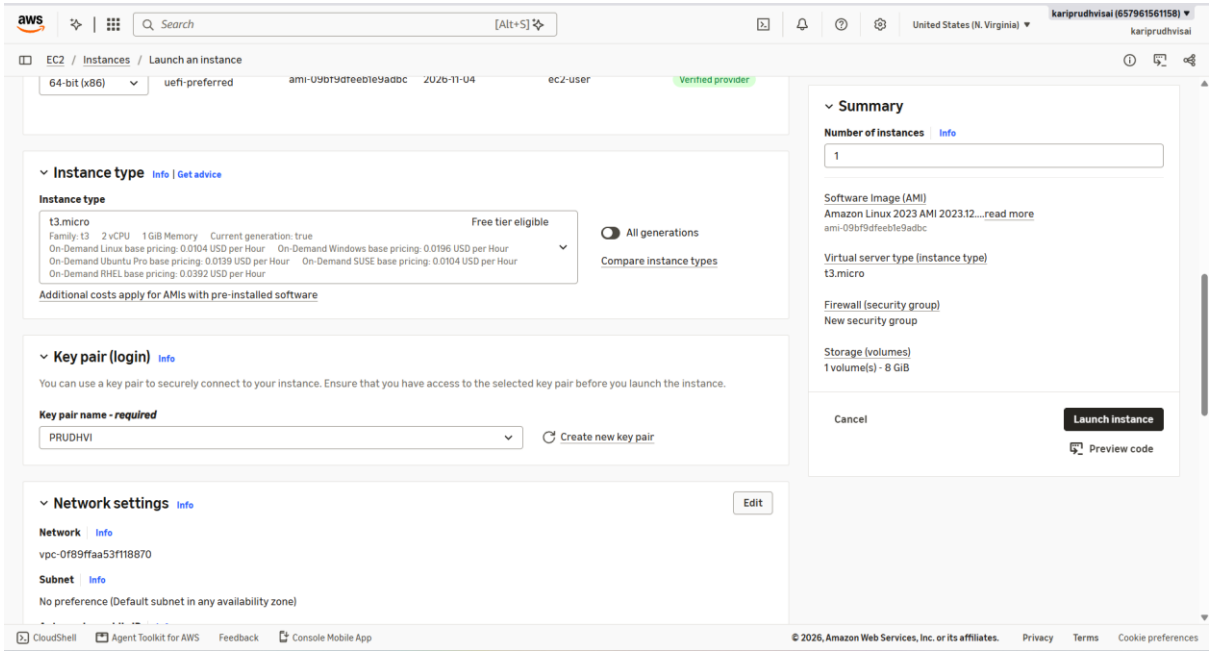
The page is scrolled down to the "Configure storage" section, showing the default configuration of a single 8 GiB gp3 root volume.



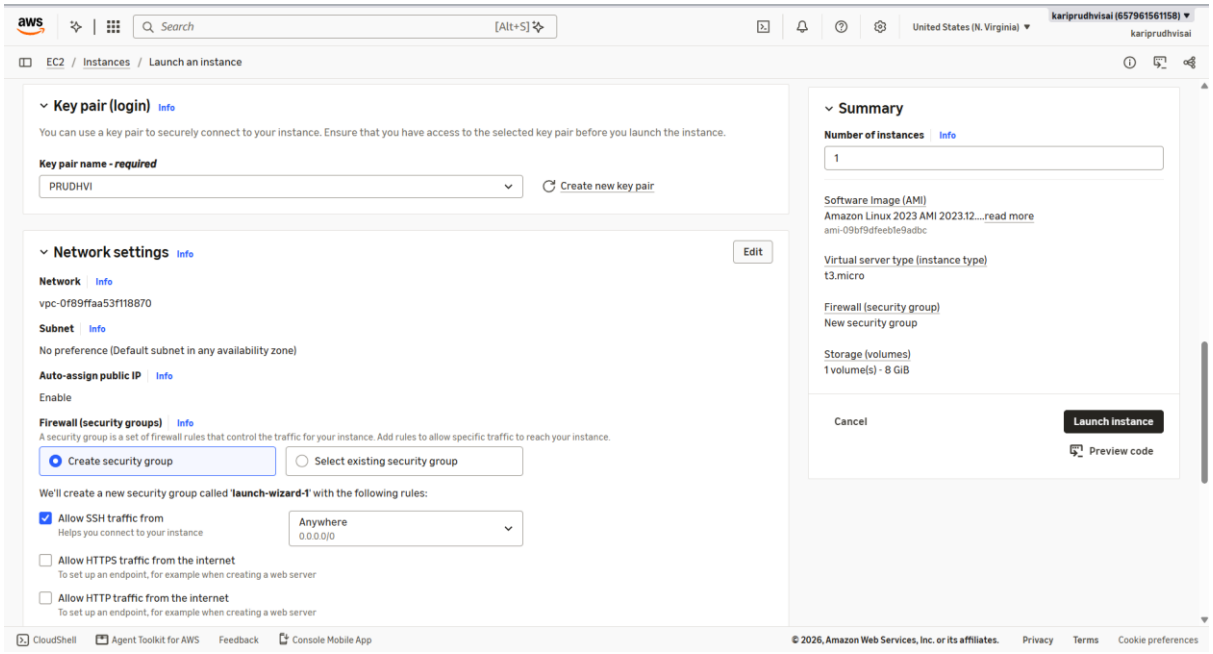
A modal dialog titled "Create key pair" is open over the instance launch page. The user is prompted to enter a "Key pair name" and the default settings of "RSA" for Key pair type and ".pem" for Private key file format are selected.



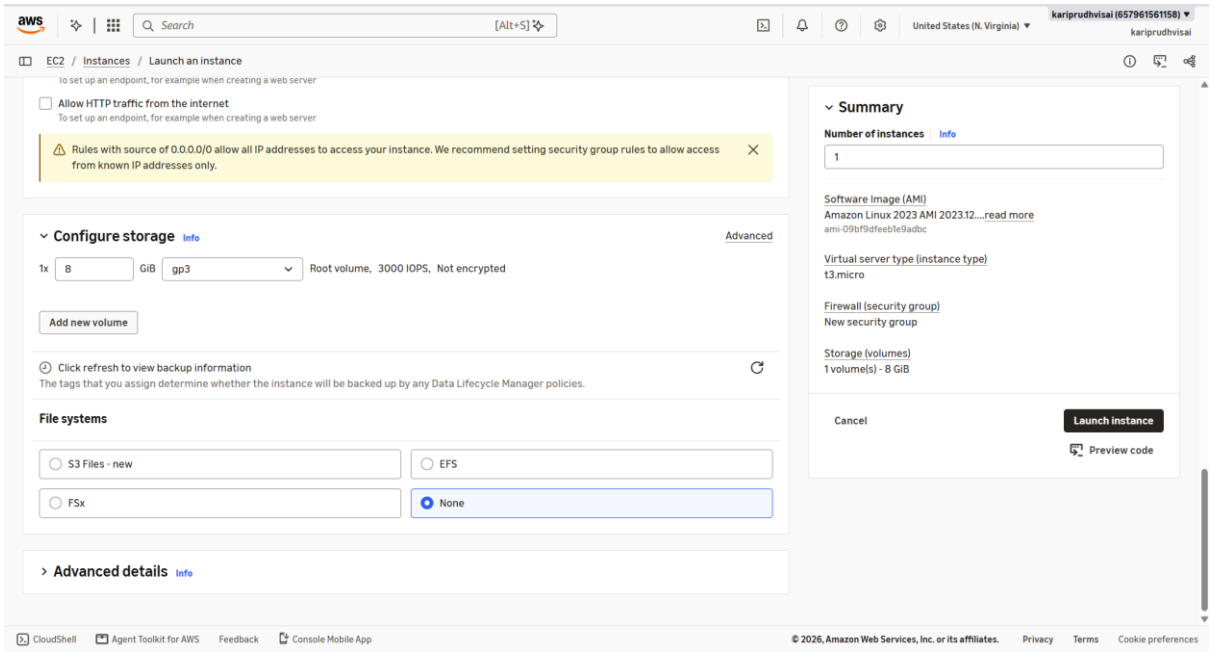
In the "Create key pair" modal, the user has entered "PRUDHVI" as the "Key pair name".



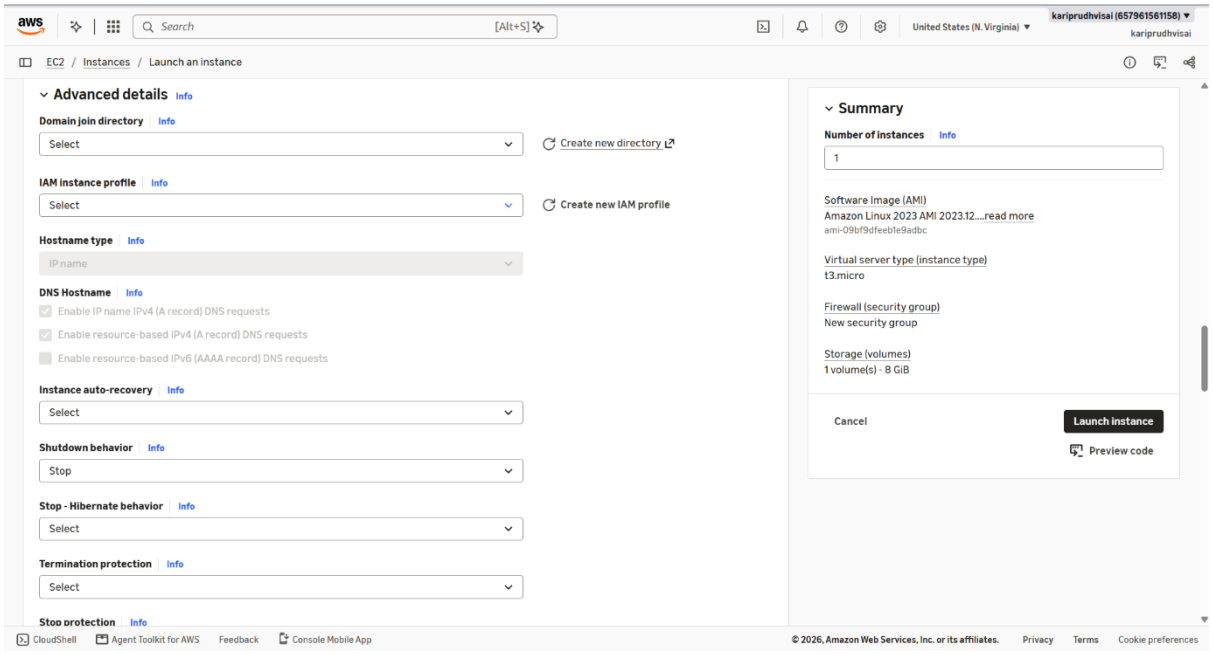
The key pair has been created, and the modal is closed. The "Key pair (login)" section now shows the newly created "PRUDHVI" key pair selected from the dropdown menu.



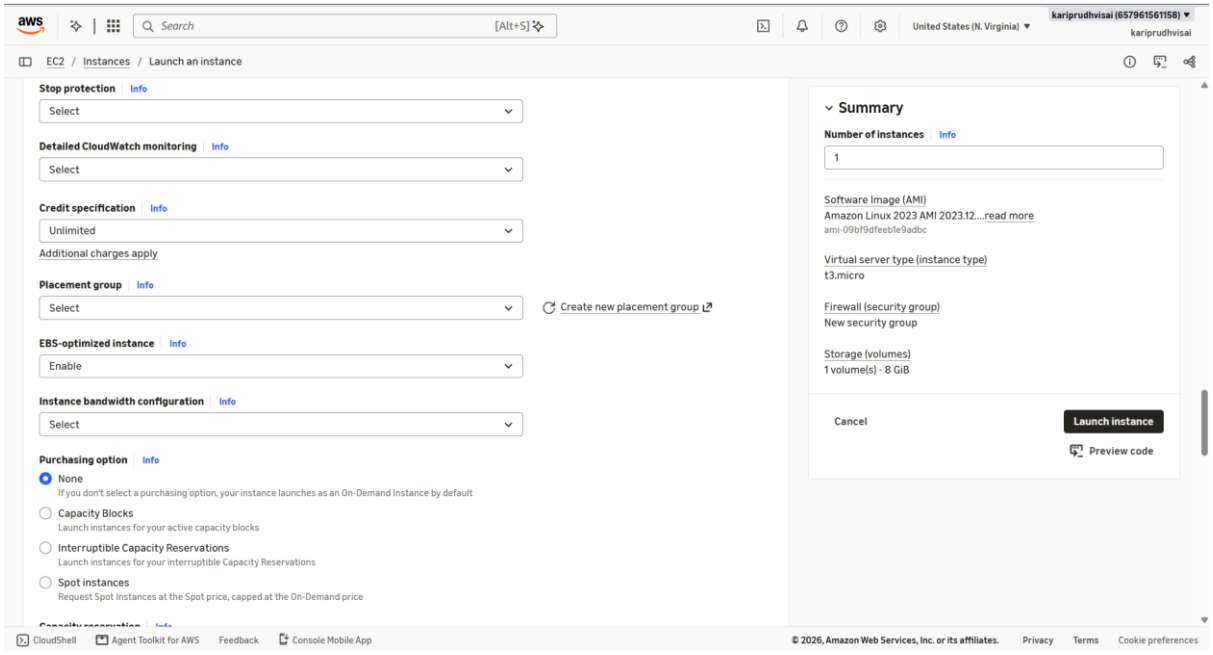
This screenshot shows the upper portion of the network settings again, confirming the "PRUDHVI" key pair is selected and the network settings remain configured to allow SSH traffic from anywhere.



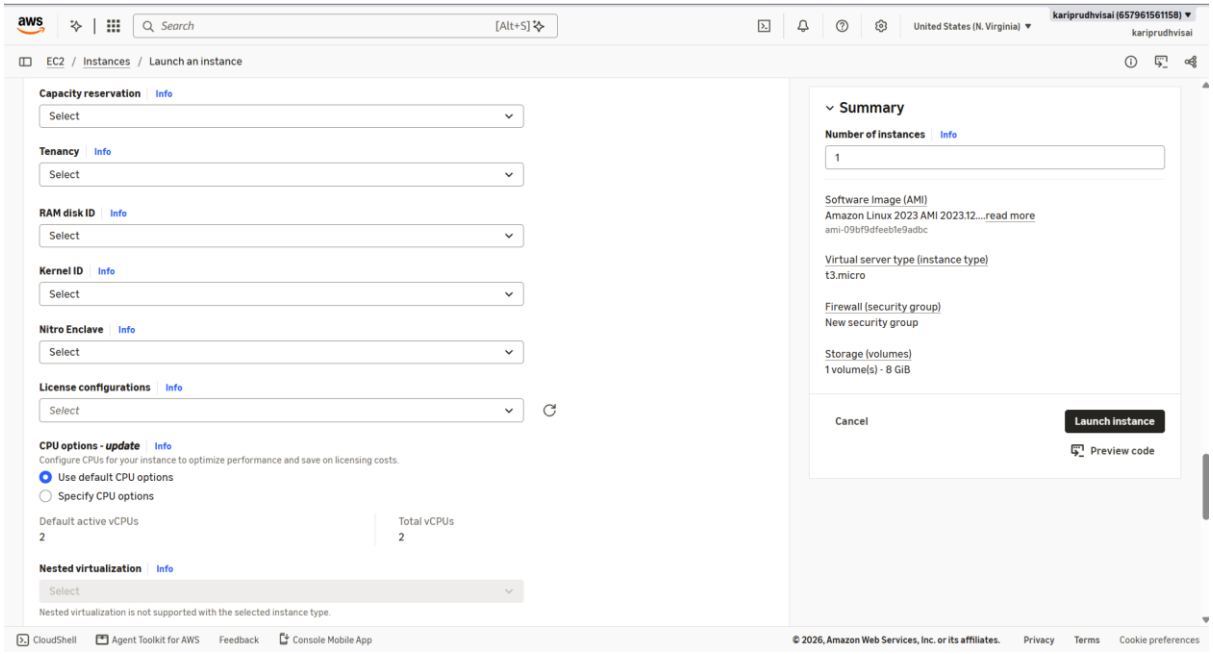
The final screenshot shows the bottom of the page again, displaying the storage configuration (8 GiB gp3) and the "Advanced details" section collapsed, with the "Launch instance" button visible on the right.



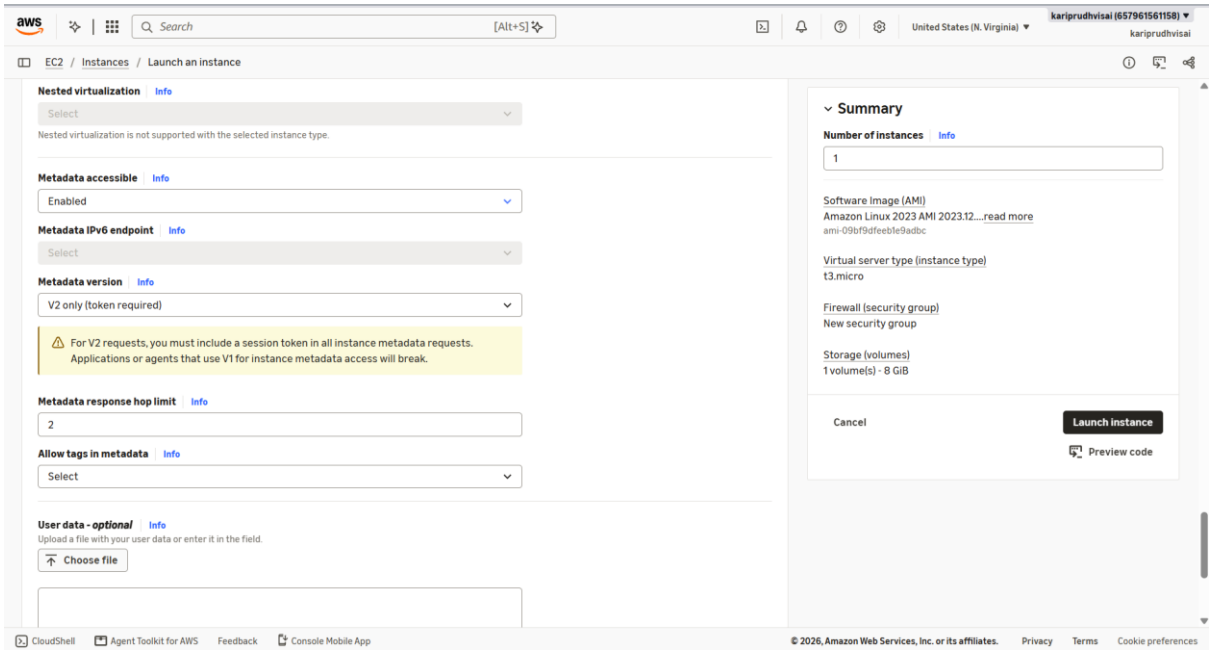
The "Advanced details" section of the "Launch an instance" page is expanded. All options, such as "Domain join directory," "IAM instance profile," and "Instance auto-recovery," are left at their default unconfigured "Select" state.



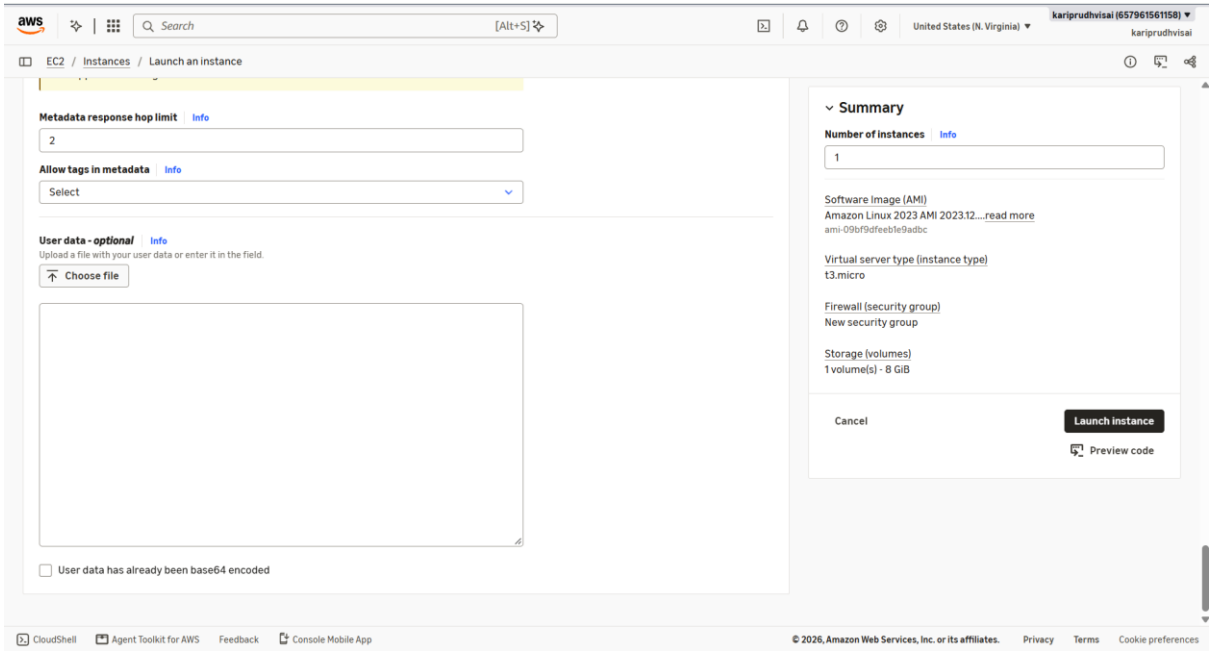
Scrolling further down in the "Advanced details" section, configurations like "Detailed CloudWatch monitoring," "Placement group," and "EBS-optimized instance" are visible. The "Purchasing option" is explicitly set to "None," meaning the instance will launch as an On-Demand Instance.



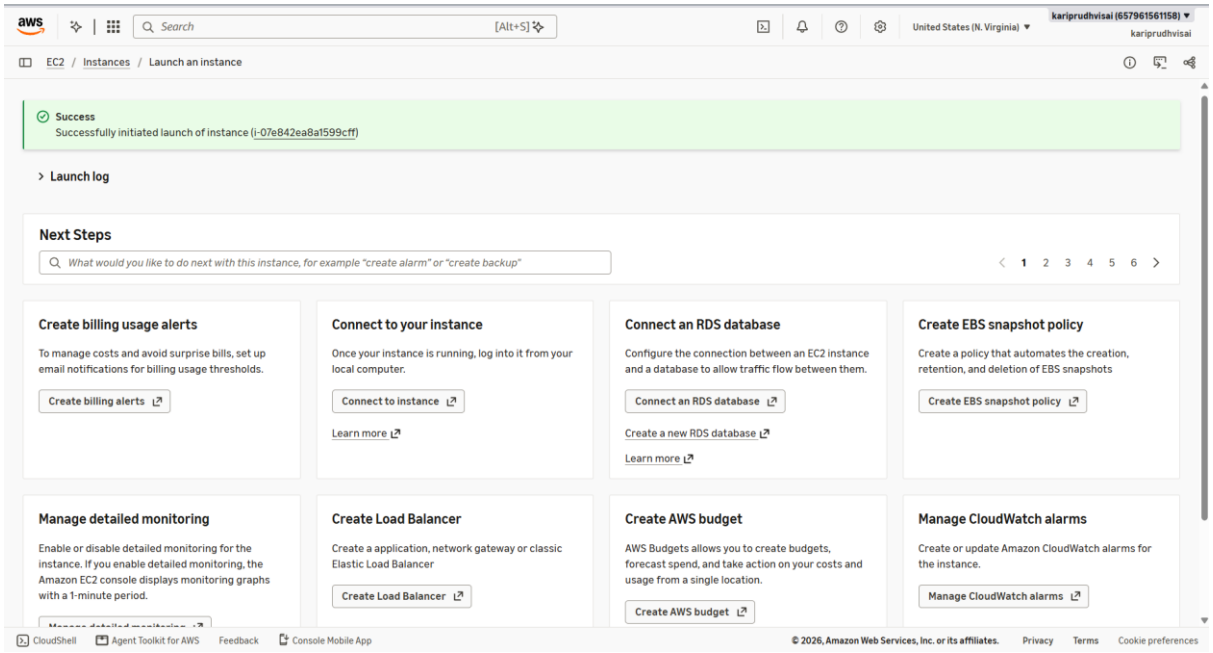
Continuing down the "Advanced details" section, options for Capacity reservation, Tenancy, and License configurations are visible in their default states. Under "CPU options," "Use default CPU options" is selected.



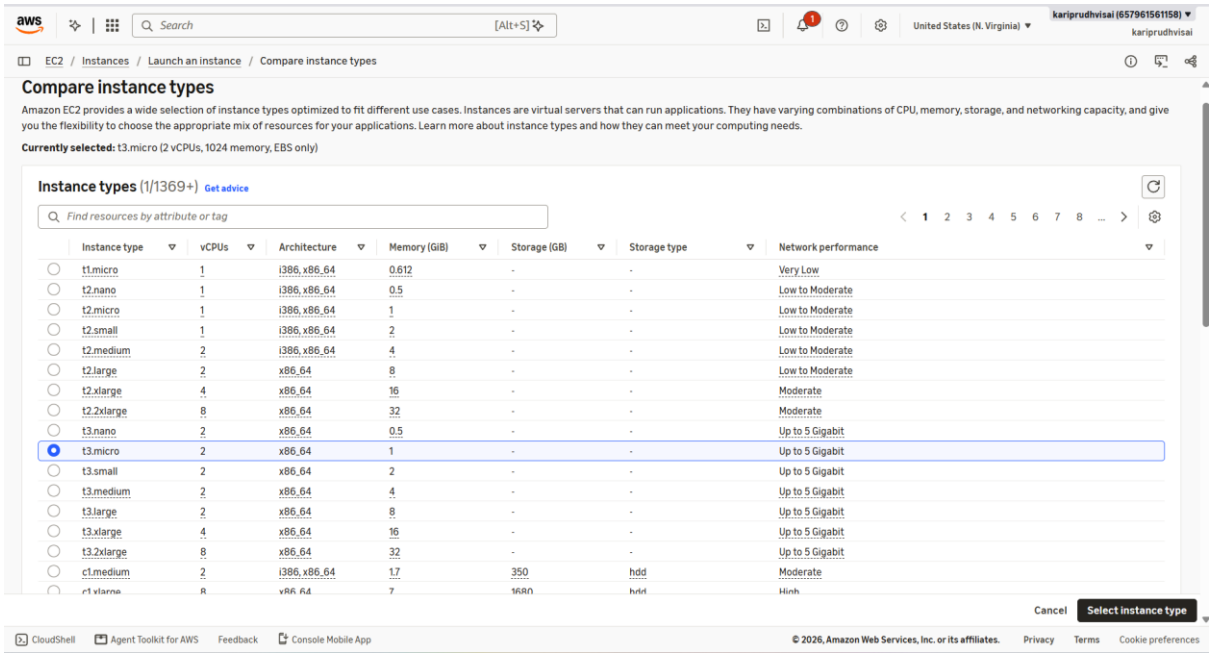
The bottom of the "Advanced details" section shows the Metadata configurations. "Metadata version" is set to "V2 only (token required)." The "User data" text field is visible but currently empty.



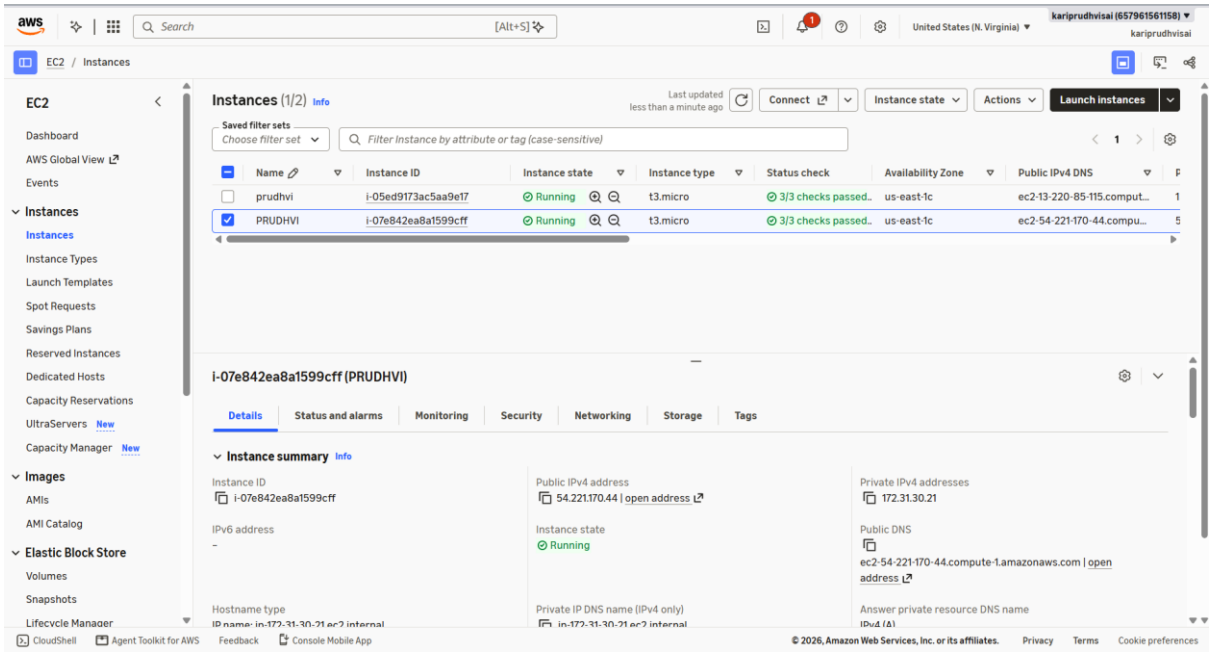
This image shows the very bottom of the "Launch an instance" page. The "User data" field remains empty. The "Launch instance" button is visible in the Summary pane on the right.



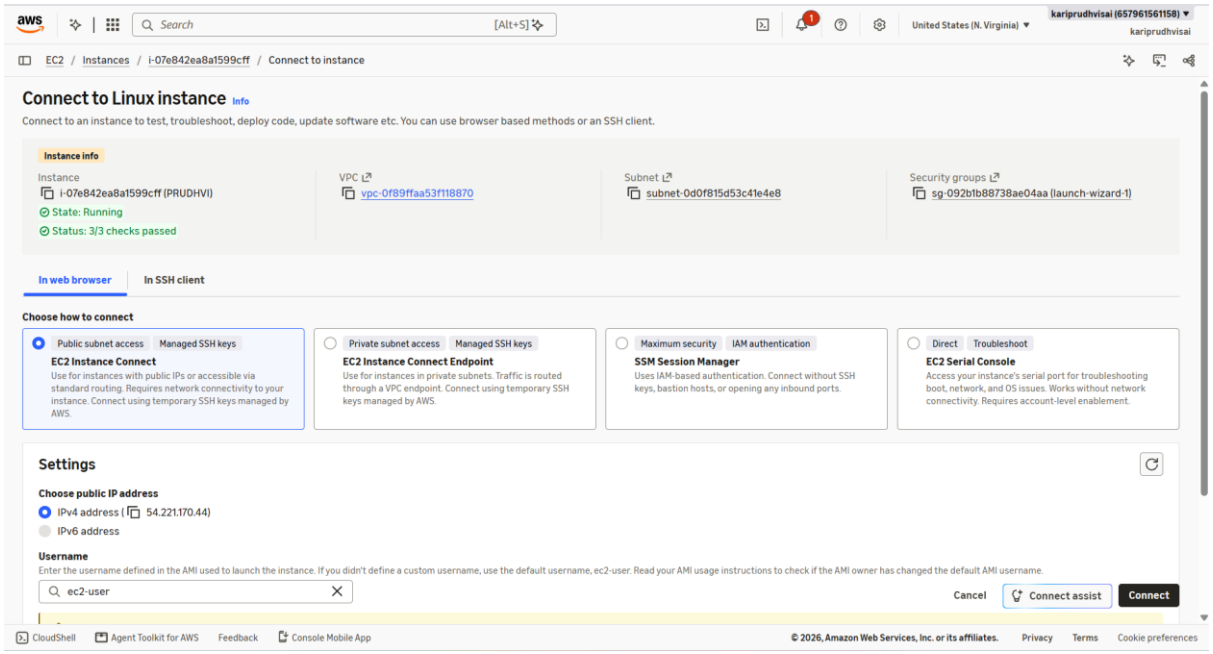
A success banner reads, "Successfully initiated launch of instance (i-07e842ea8a1599cff)." Below it, a "Next Steps" section offers shortcuts to actions like creating billing alerts, connecting to the instance, or managing CloudWatch alarms.



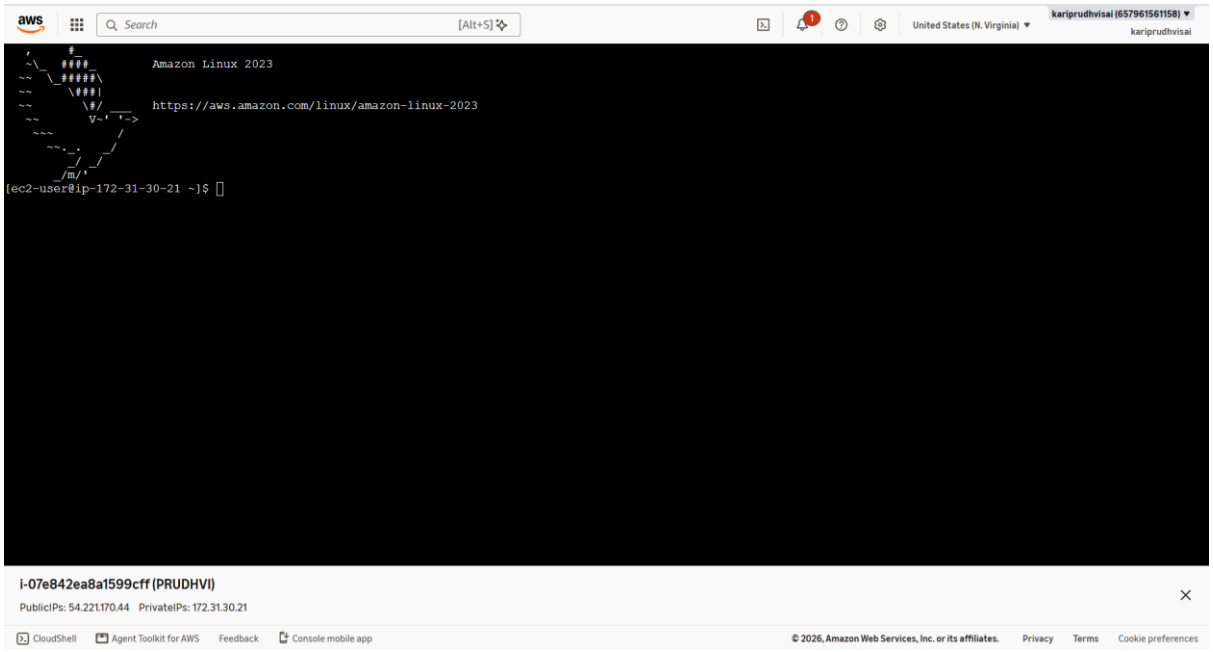
A "Compare instance types" window is open, displaying a list of available EC2 instance types. The "t3.micro" instance type is currently selected and highlighted, showing its specifications: 2 vCPUs, x86_64 architecture, and 1 GiB of memory.



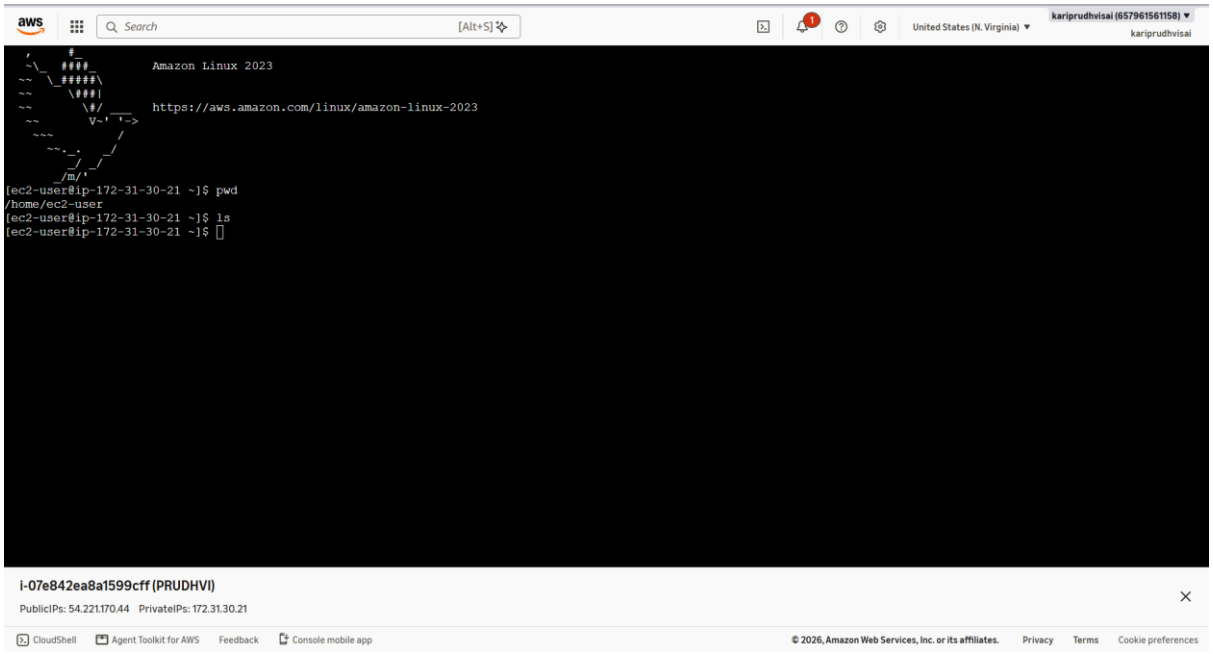
The EC2 Instances dashboard now shows the newly created instance named "PRUDHVI" with the ID "i-07e842ea8a1599cff". Its "Instance state" is "Running," and its "Status check" shows "3/3 checks passed." The Details tab is open in the lower pane, displaying the Public IPv4 address "54.221.170.44".



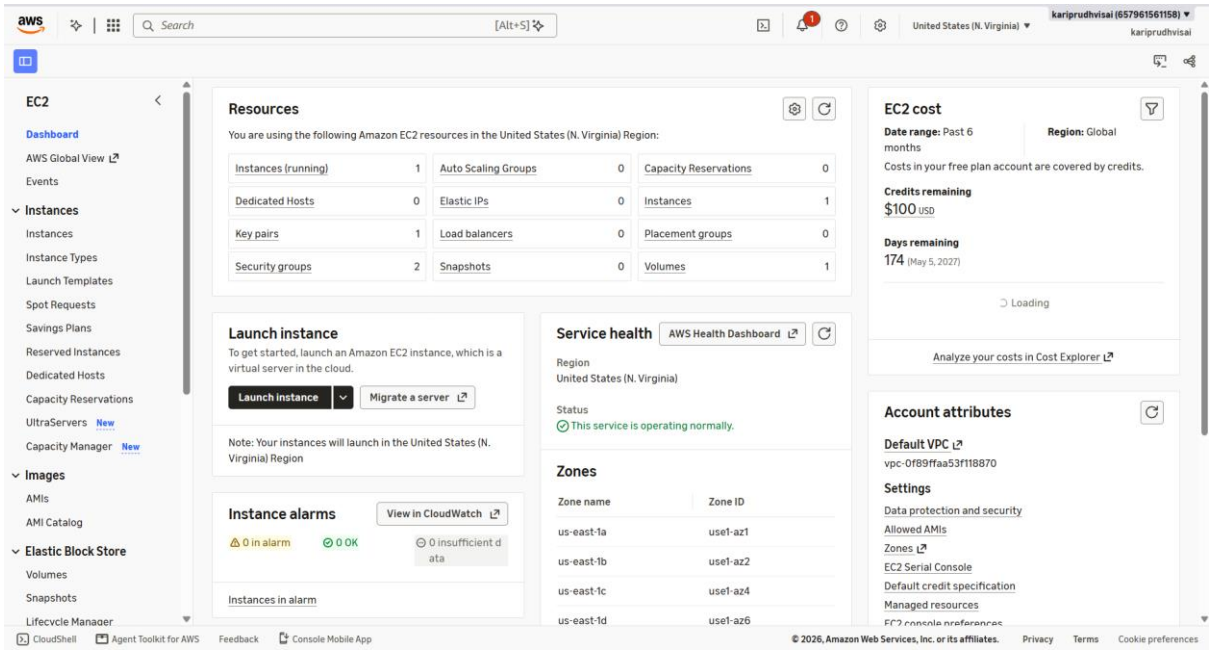
The "Connect to Linux instance" page is displayed for the "PRUDHVI" instance. The "In web browser" tab is selected, and the "EC2 Instance Connect" method is chosen. The "Username" field is populated with "ec2-user".



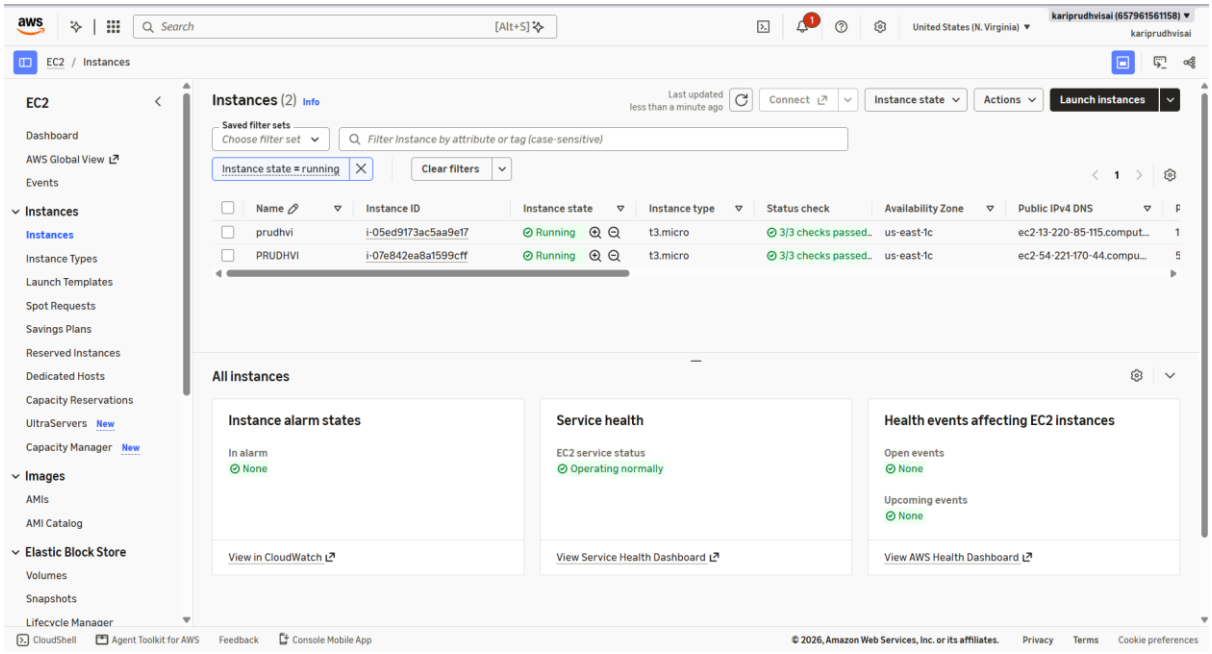
This screenshot shows an active terminal session connected to the "PRUDHVI" instance via the web-based EC2 Instance Connect. The command prompt indicates a successful login as ec2-user@ip-172-31-30-21, and the Amazon Linux 2023 ASCII art logo is displayed.



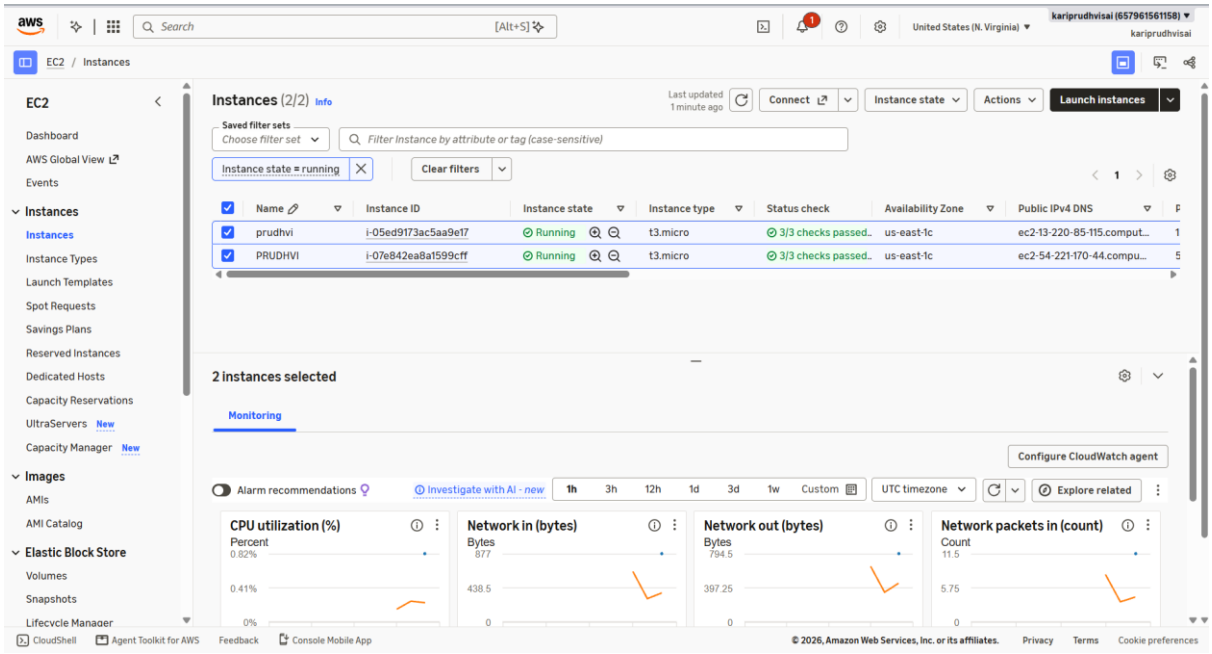
The web-based terminal session is active. The pwd command is executed, returning /home/ec2-user, followed by the ls command, which returns an empty prompt, indicating no files are in the directory.



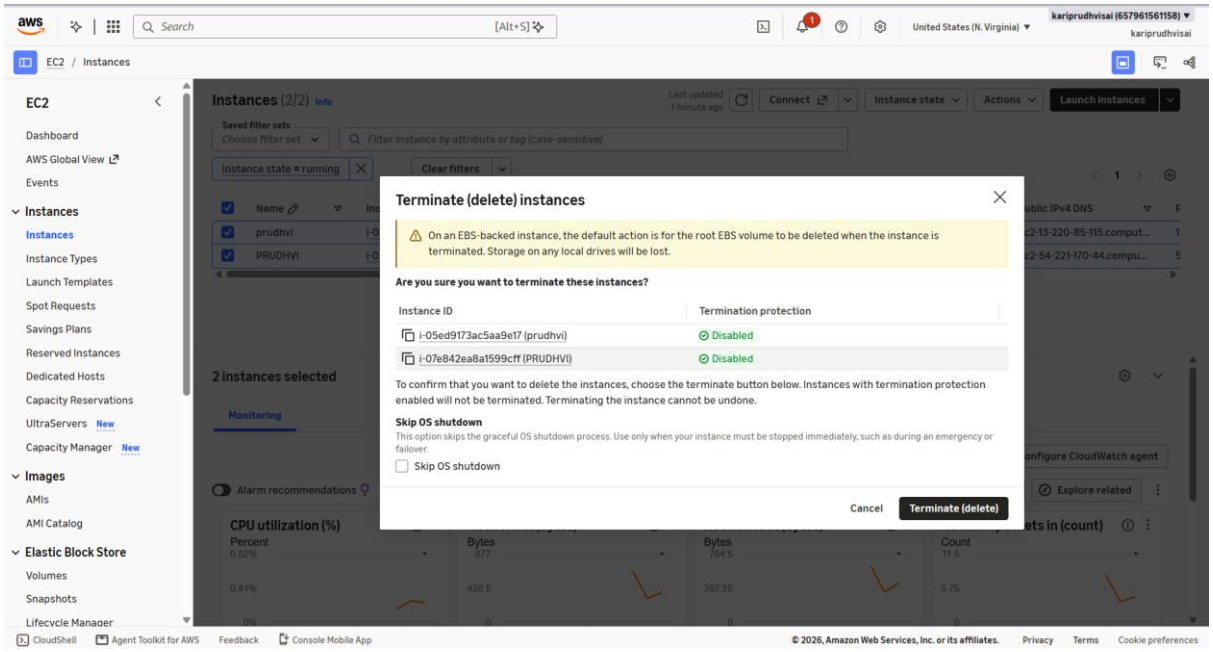
The EC2 Dashboard shows the updated resource counts, with 1 Instance (running), 1 Key pair, 2 Security groups, and 1 Volume now active in the N. Virginia region.



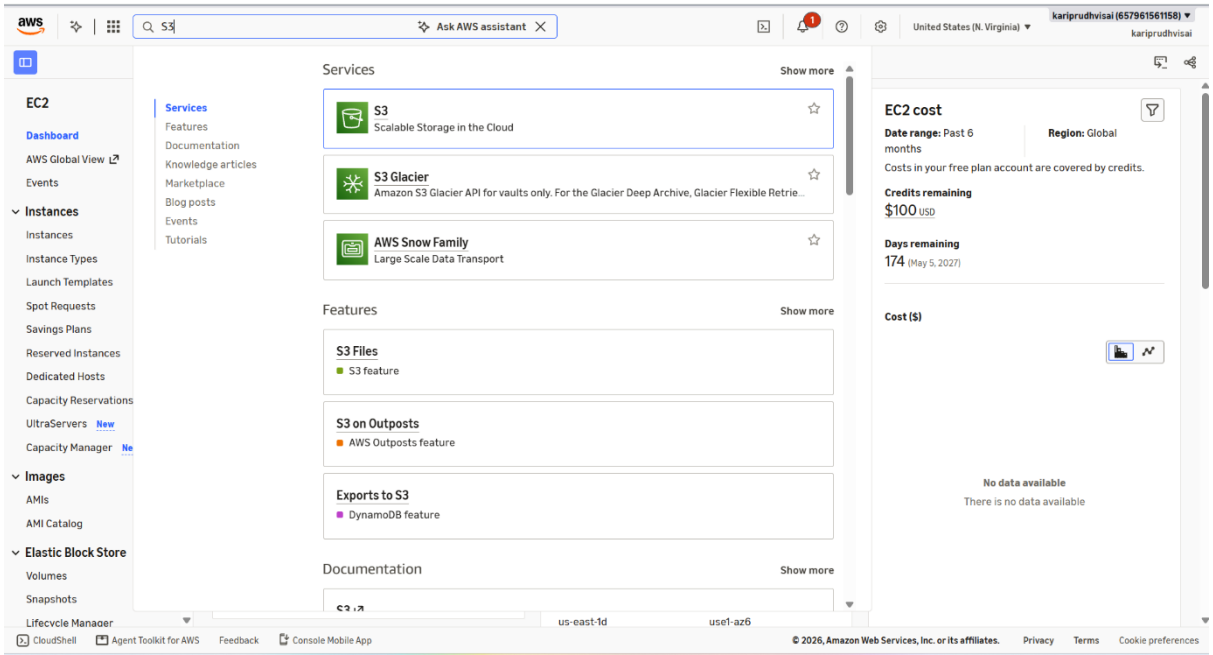
The Instances dashboard is filtered by Instance state = running, showing two instances: "prudhvi" (i-05ed...) and the newly created "PRUDHVI" (i-07e8...). Both are t3.micro instances and have passed their status checks.



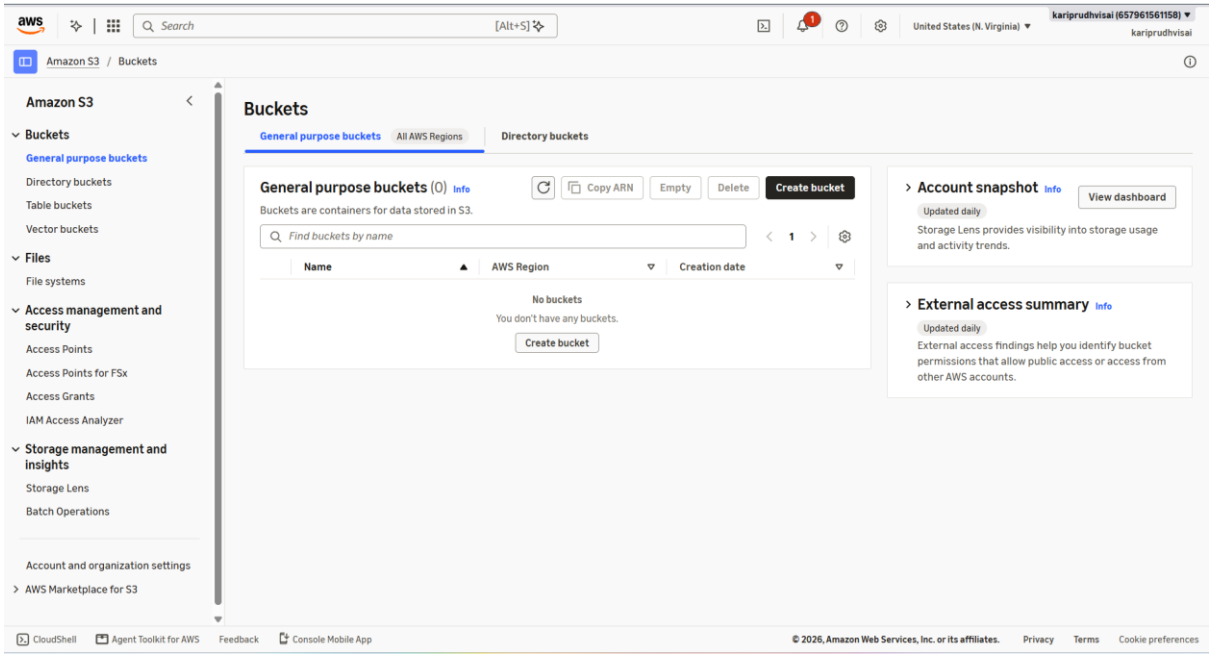
Both instances are selected on the Instances dashboard. The lower pane displays the "Monitoring" tab for the "2 instances selected," showing aggregate CloudWatch metrics for CPU utilization, Network in, Network out, and Network packets in.



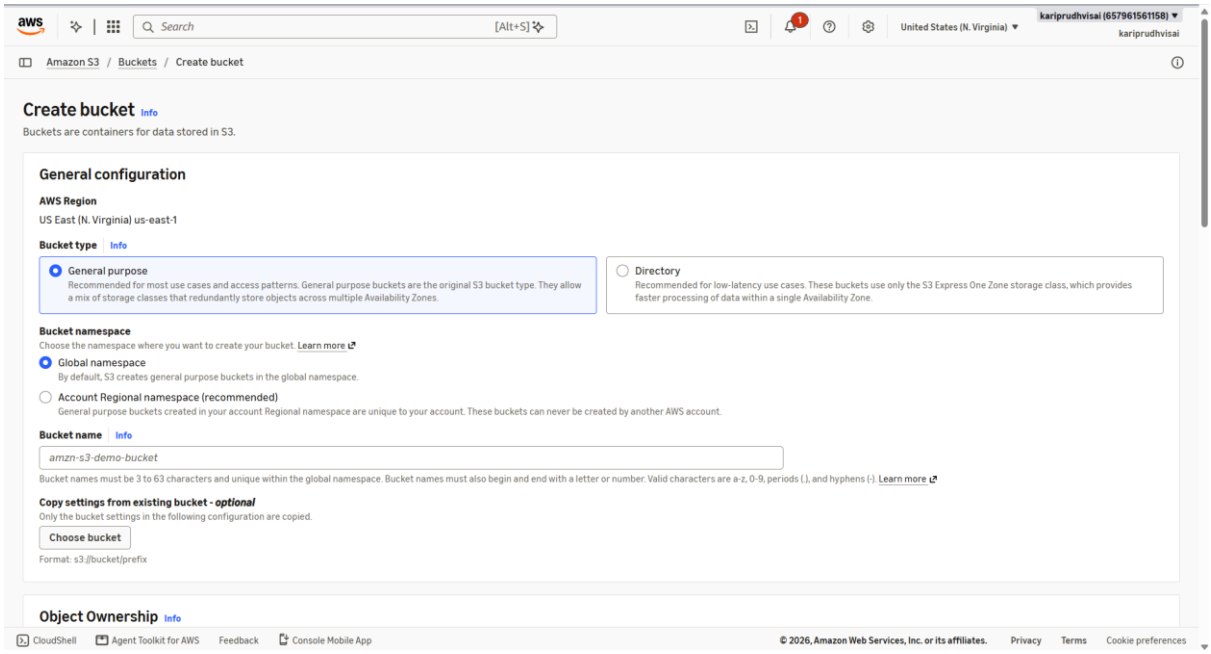
A "Terminate (delete) instances" modal dialog confirms the termination of both selected instances. The warning states that the root EBS volumes will be deleted.



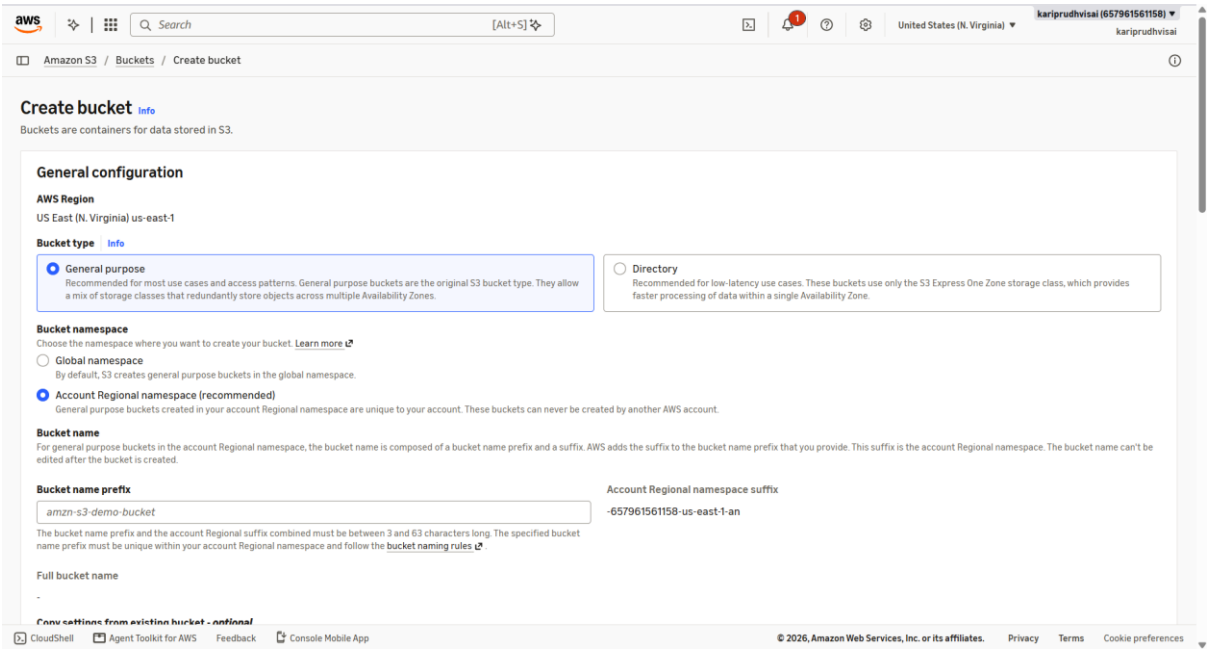
The unified search bar is used to search for "S3". The top result is the "S3 (Scalable Storage in the Cloud)" service.



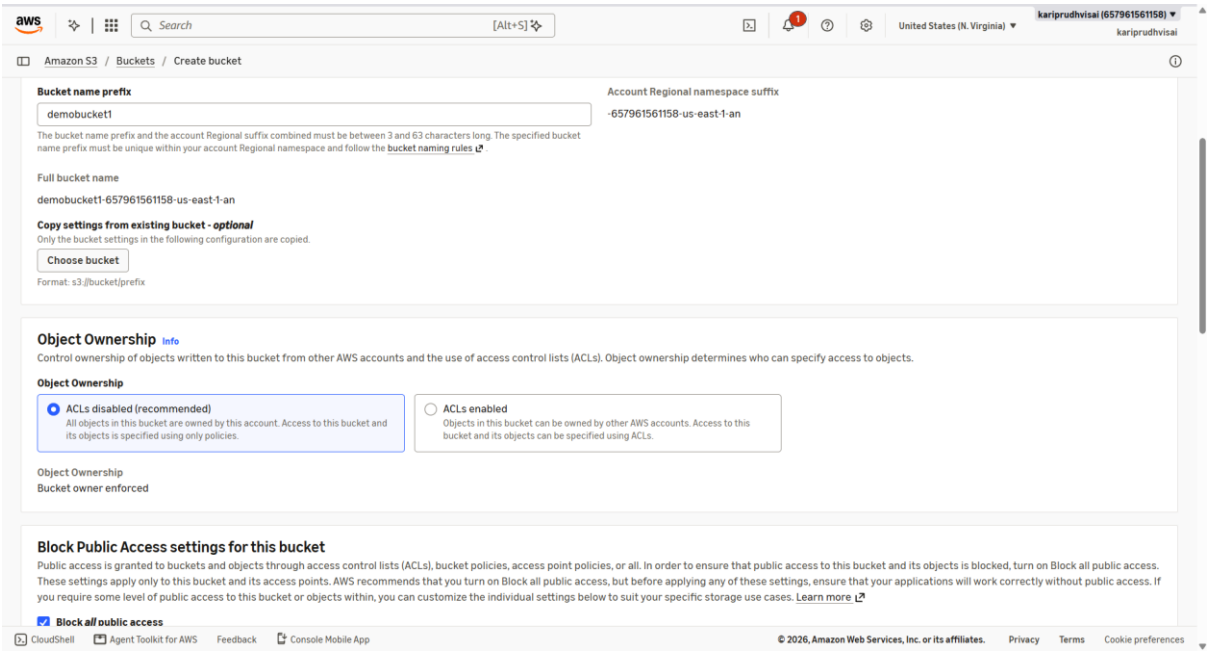
The Amazon S3 Buckets dashboard shows "0" General purpose buckets, indicating the account currently has no buckets created.



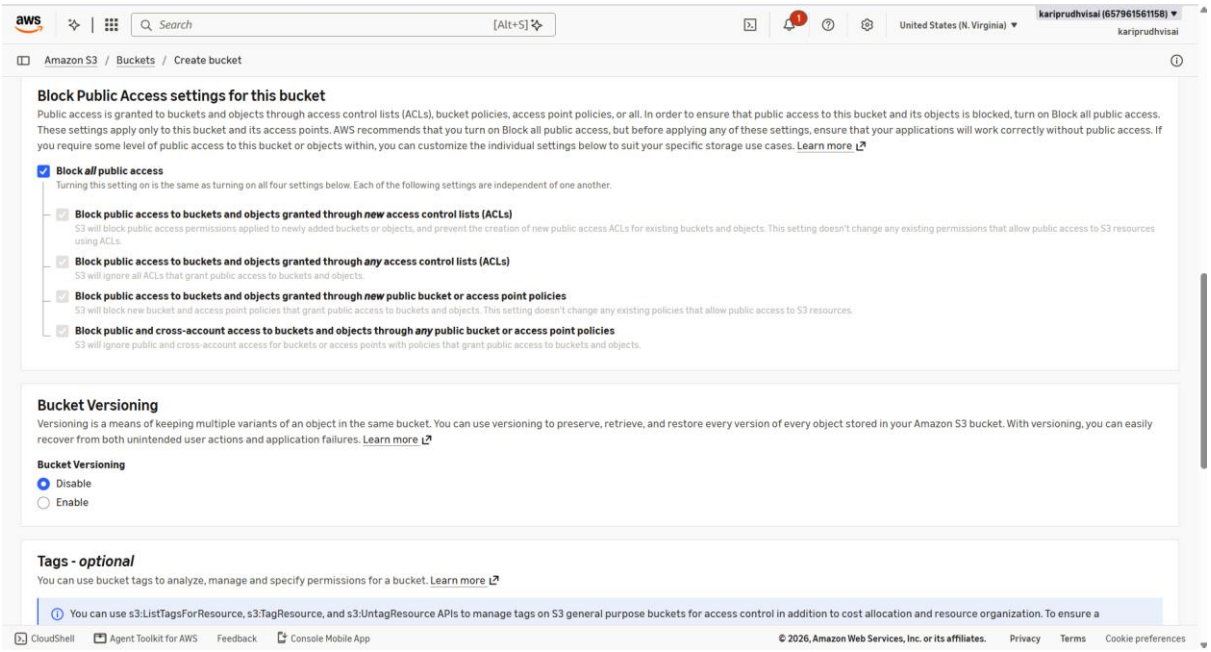
The "Create bucket" page is open. The "General purpose" bucket type and "Global namespace" are selected. The "Bucket name" field contains placeholder text.



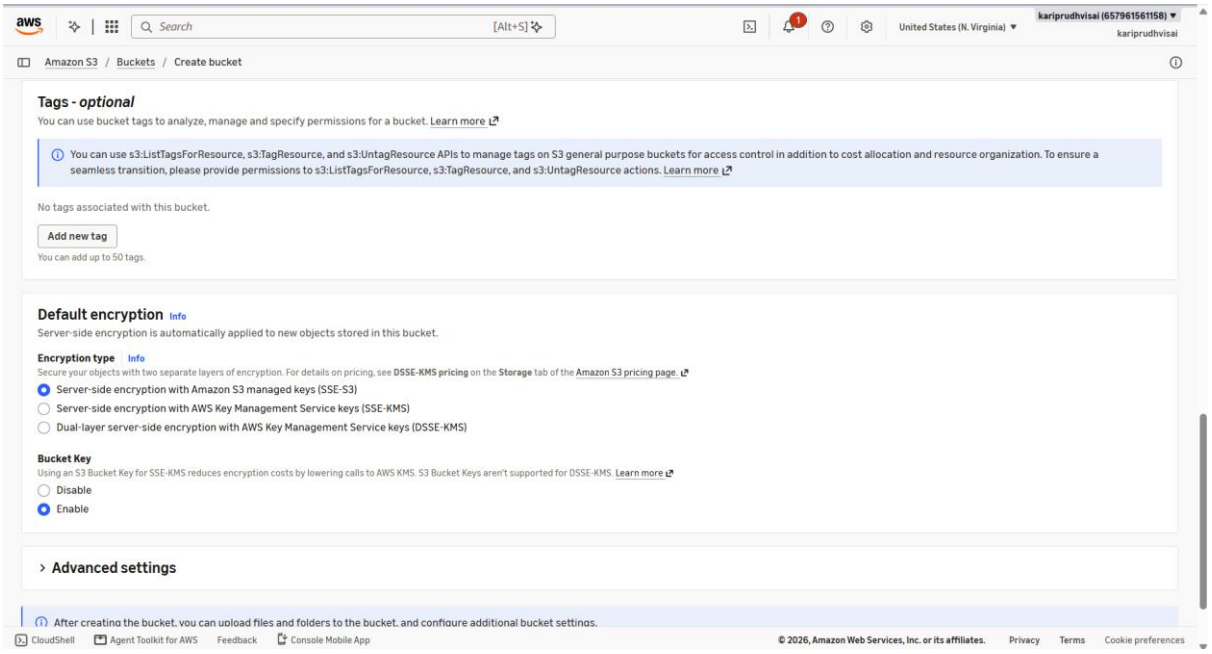
On the "Create bucket" page, the user has changed the "Bucket namespace" selection to "Account Regional namespace (recommended)". This adds an "Account Regional namespace suffix" (-657961561158-us-east-1-an) to the bucket naming convention.



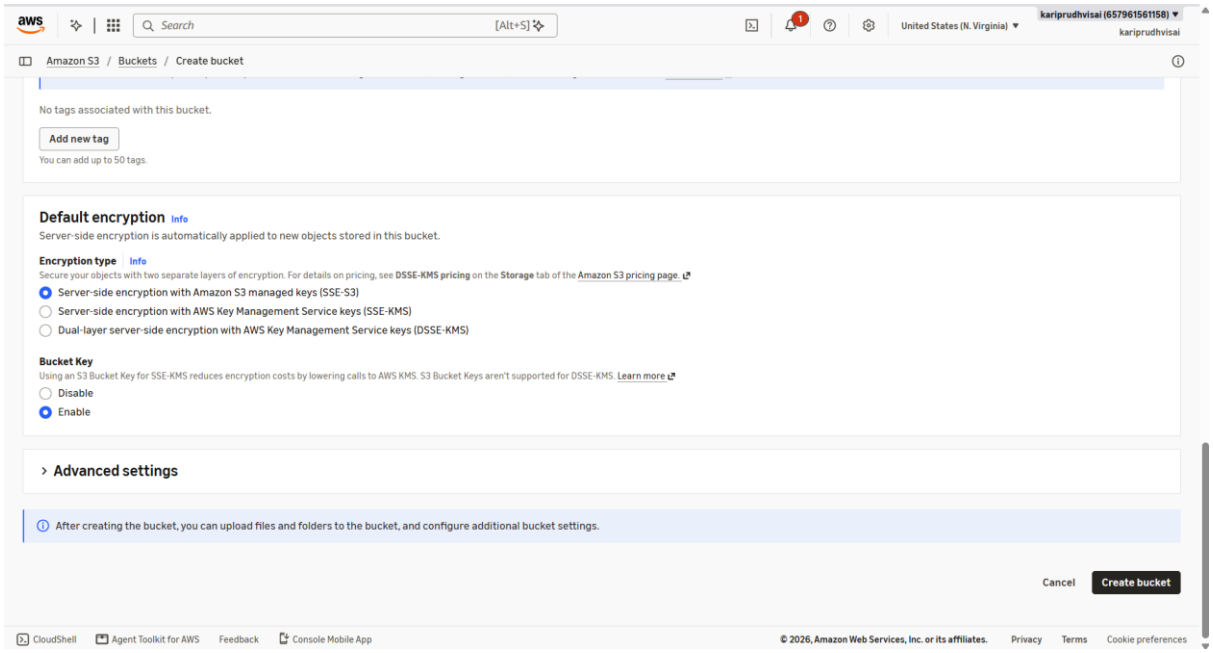
The user has entered "demobucket1" as the "Bucket name prefix". The "Full bucket name" is generated as demobucket1-657961561158-us-east-1-an. Below, the "Object Ownership" setting is set to the recommended "ACLs disabled".



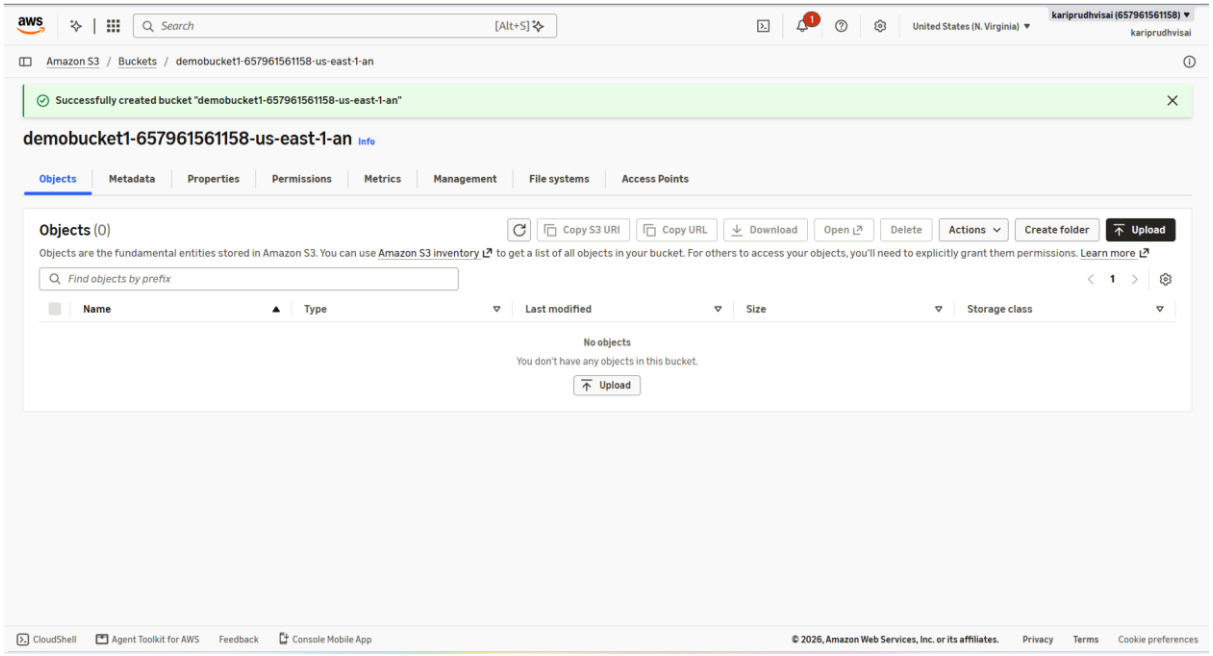
A continuation of the "Create bucket" page shows the "Object Ownership" setting explicitly configured to "ACLs disabled (recommended)". Below this, the "Block Public Access settings for this bucket" section has the "Block all public access" checkbox enabled.



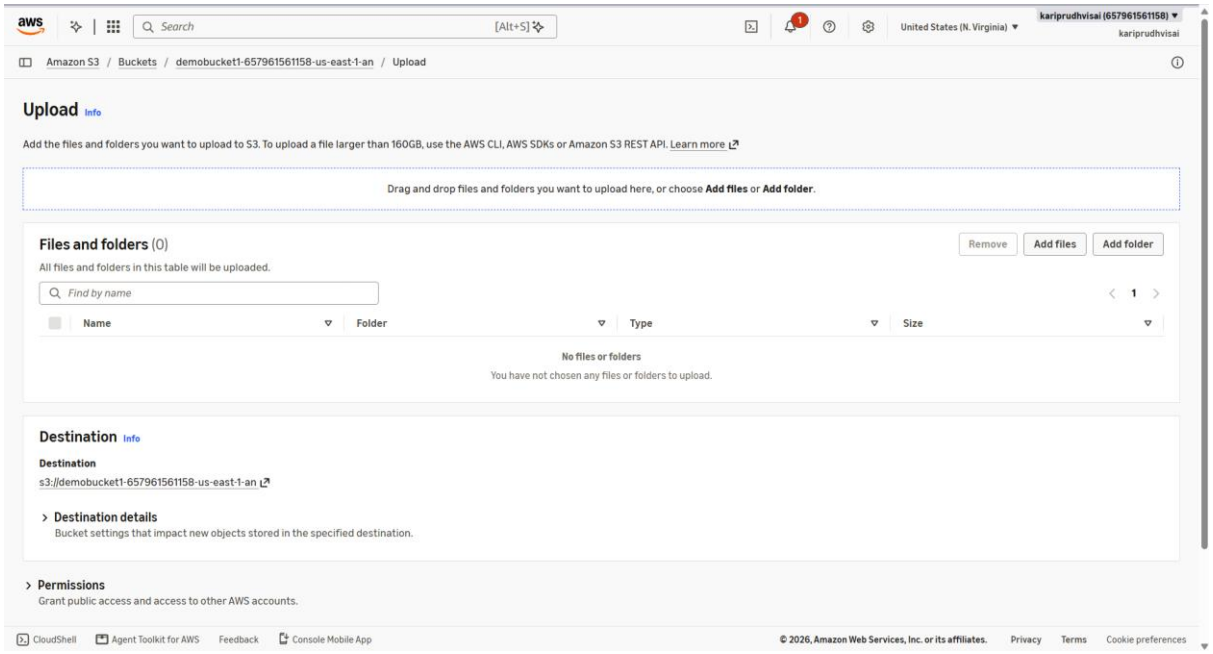
Scrolling down the "Create bucket" page, the detailed breakdown of the four individual settings within "Block all public access" is visible, all checked. Below this, the "Bucket Versioning" feature is set to its default "Disable" state.



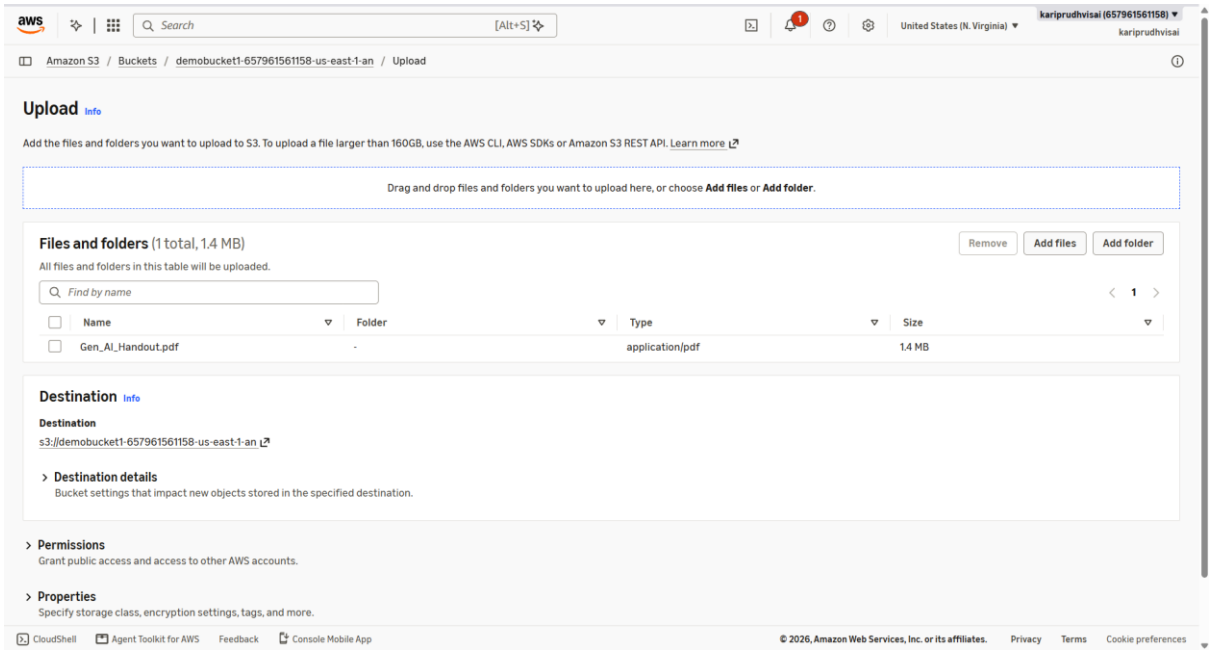
In the "Tags" section, no tags are currently associated with the new bucket. The "Default encryption" is set to "Server-side encryption with Amazon S3 managed keys (SSE-S3)" and "Bucket Key" is enabled.



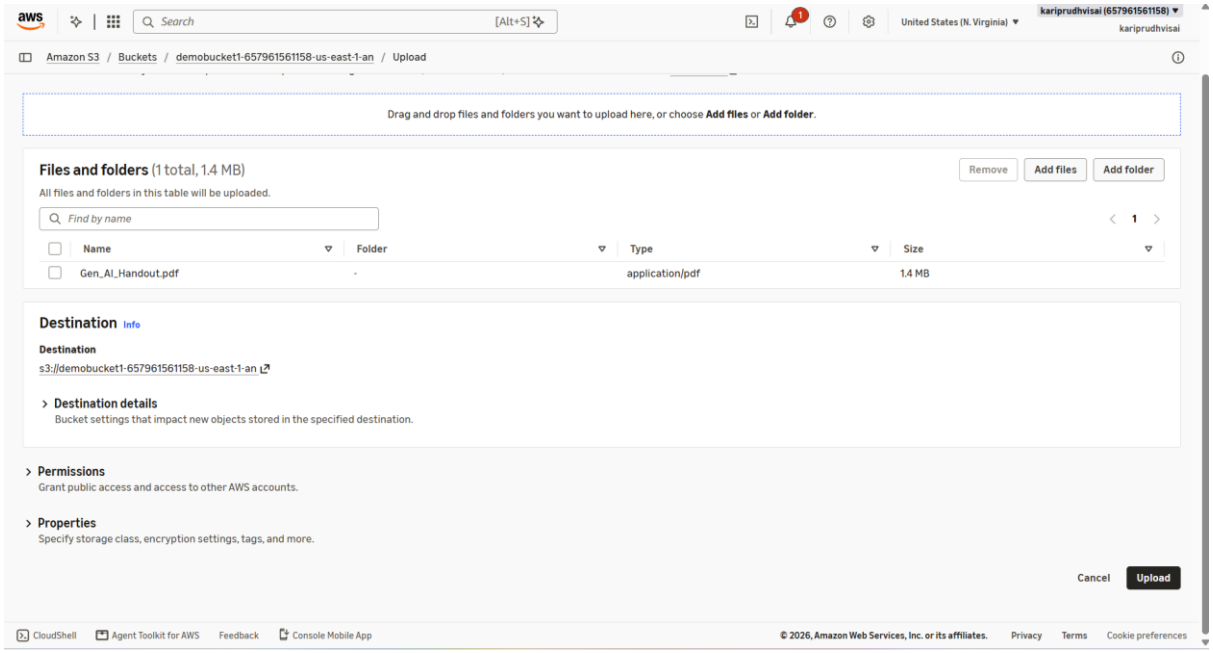
The very bottom of the "Create bucket" page shows the "Advanced settings" section collapsed, with the "Create bucket" button visible in the bottom right corner.



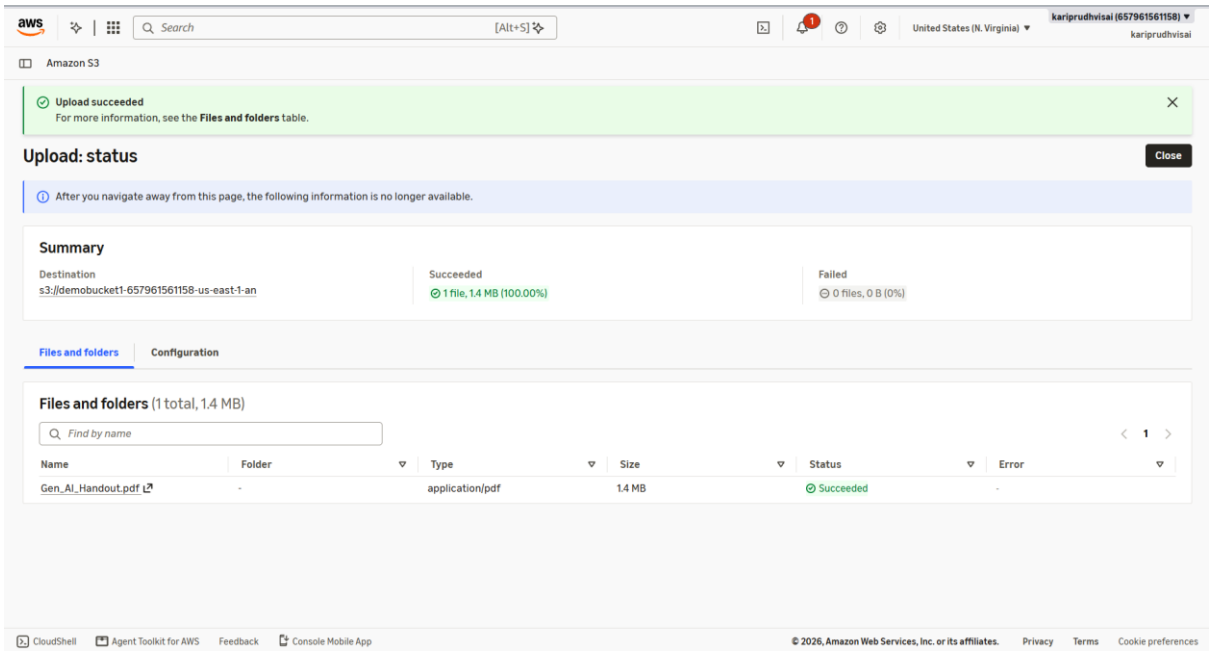
A success banner confirms, "Successfully created bucket 'demobucket1-657961561158-us-east-1-an'". The bucket's "Objects" tab is displayed, showing "No objects" currently in the bucket, with a prominent "Upload" button.



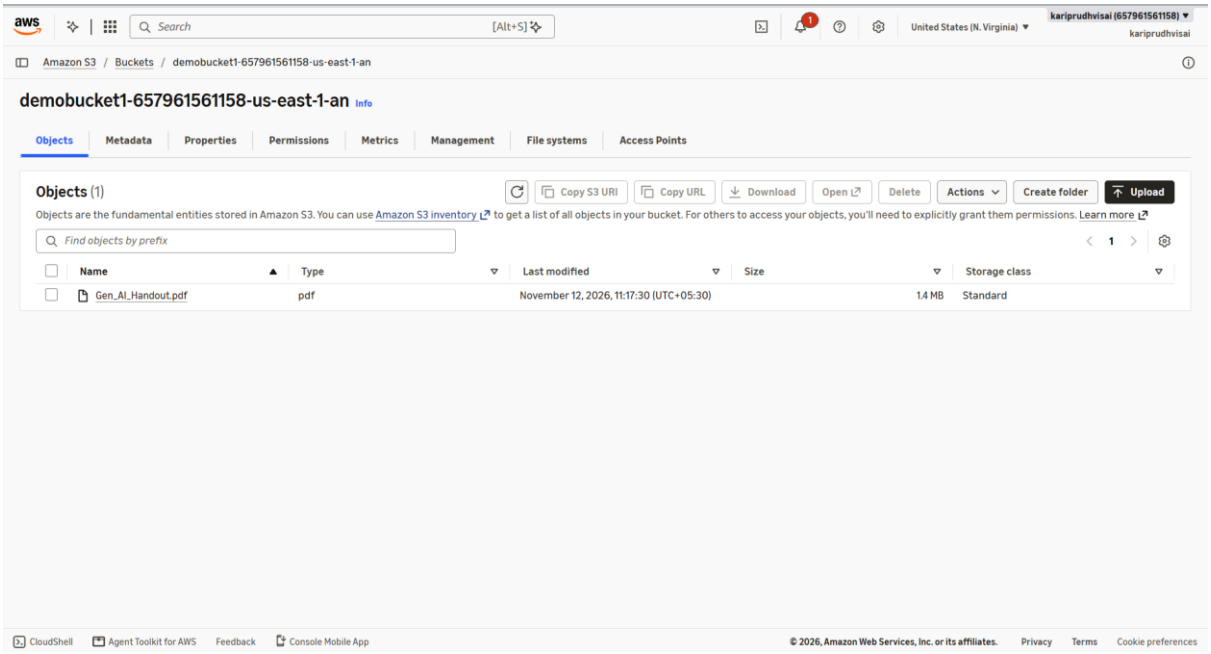
The "Upload" page for the newly created S3 bucket is open. The "Files and folders" list is currently empty, instructing the user to drag and drop files or use the "Add files" or "Add folder" buttons.



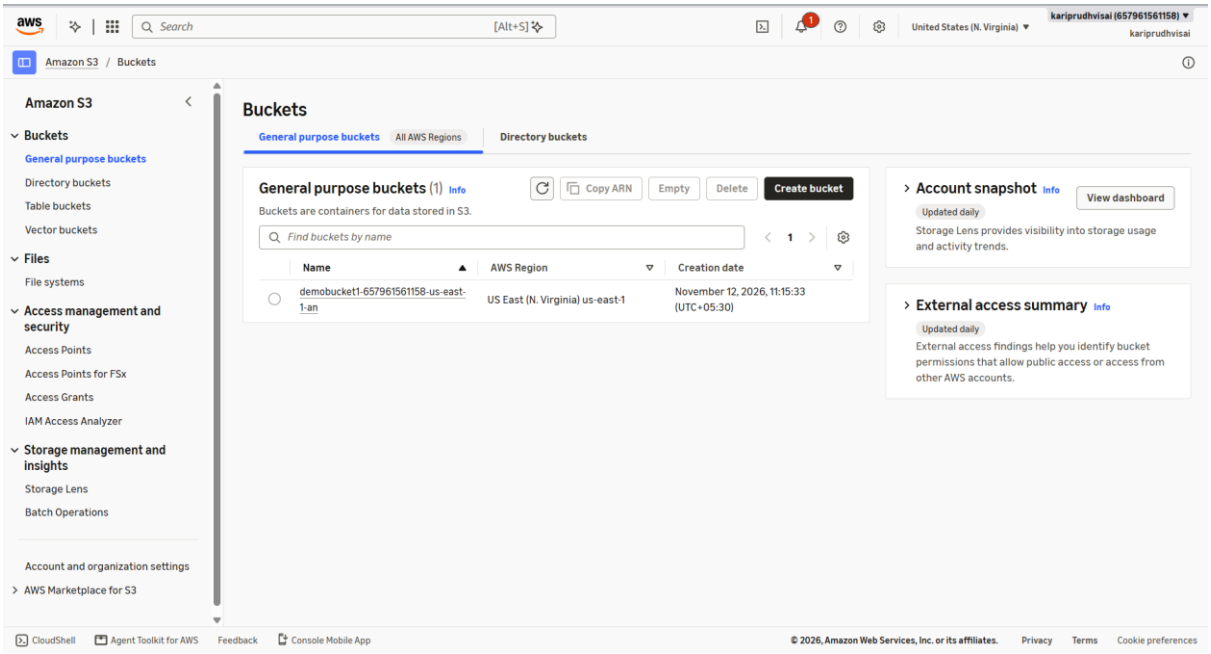
A file named Gen_AI_Handout.pdf has been added to the "Files and folders" list for upload. The list shows it is an application/pdf file with a size of 1.4 MB.



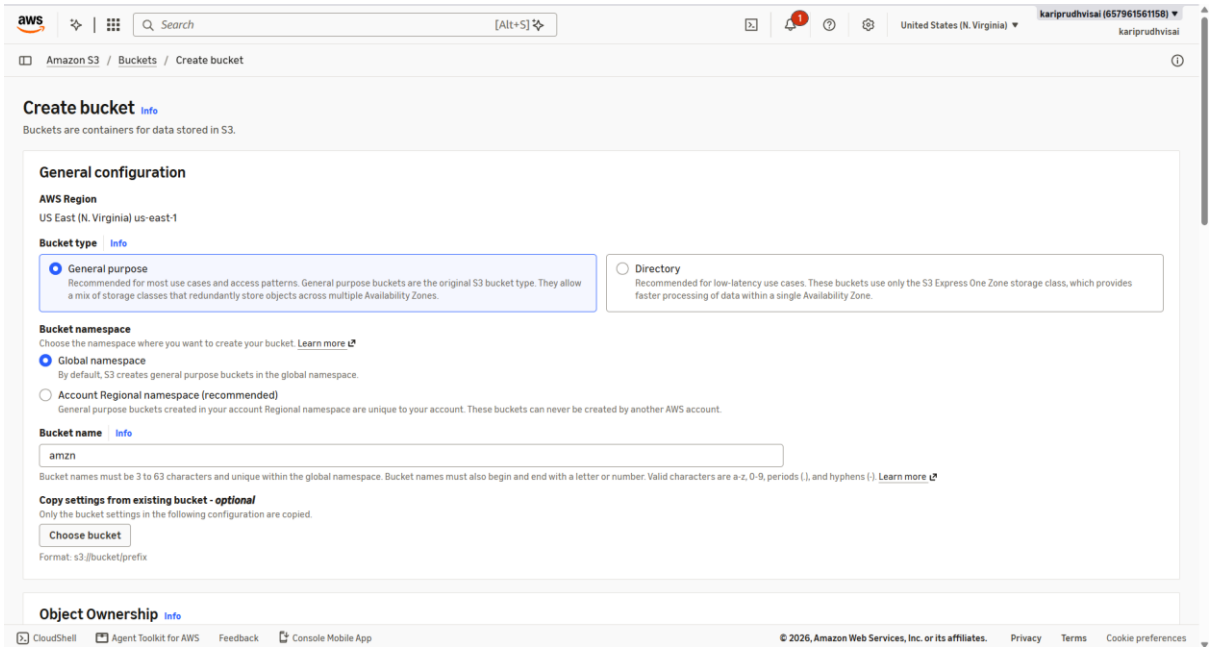
This screenshot is nearly identical to the previous one, showing the Gen_AI_Handout.pdf file queued for upload, but scrolled slightly down to reveal the "Upload" button at the bottom right.



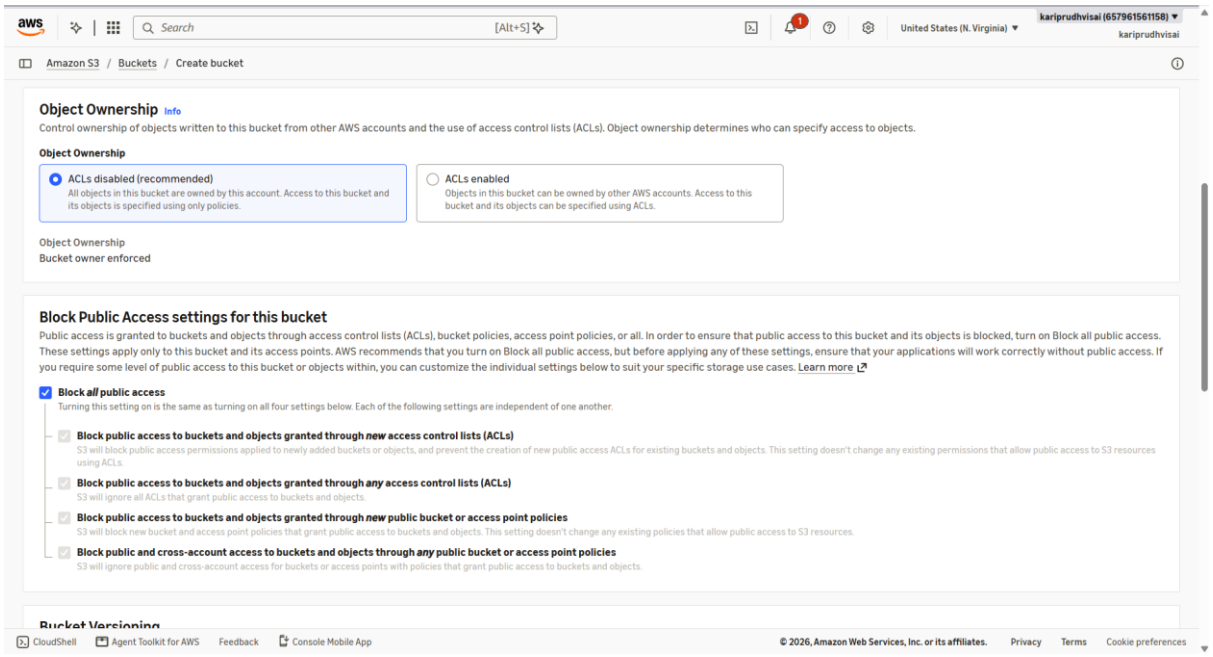
An "Upload succeeded" banner is displayed. The "Upload: status" summary confirms that 1 file (1.4 MB) succeeded and 0 failed. The detailed list below shows Gen_AI_Handout.pdf with a green "Succeeded" status.



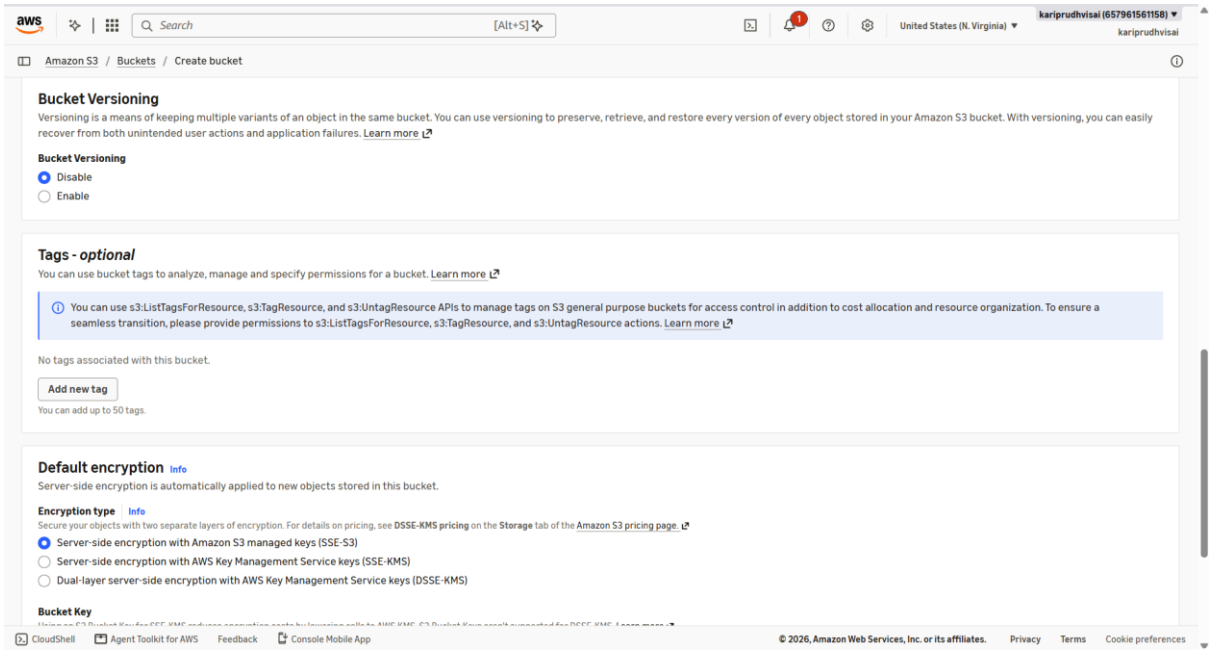
The bucket's "Objects" dashboard now lists the uploaded Gen_AI_Handout.pdf file. It shows a Last modified timestamp of November 12, 2026, a size of 1.4 MB, and a Storage class of "Standard". The Amazon S3 Buckets dashboard shows a single General purpose bucket listed: demobucket1-657961561158-us-east-1-an. It shows a creation date of November 12, 2026, 11:15:33.



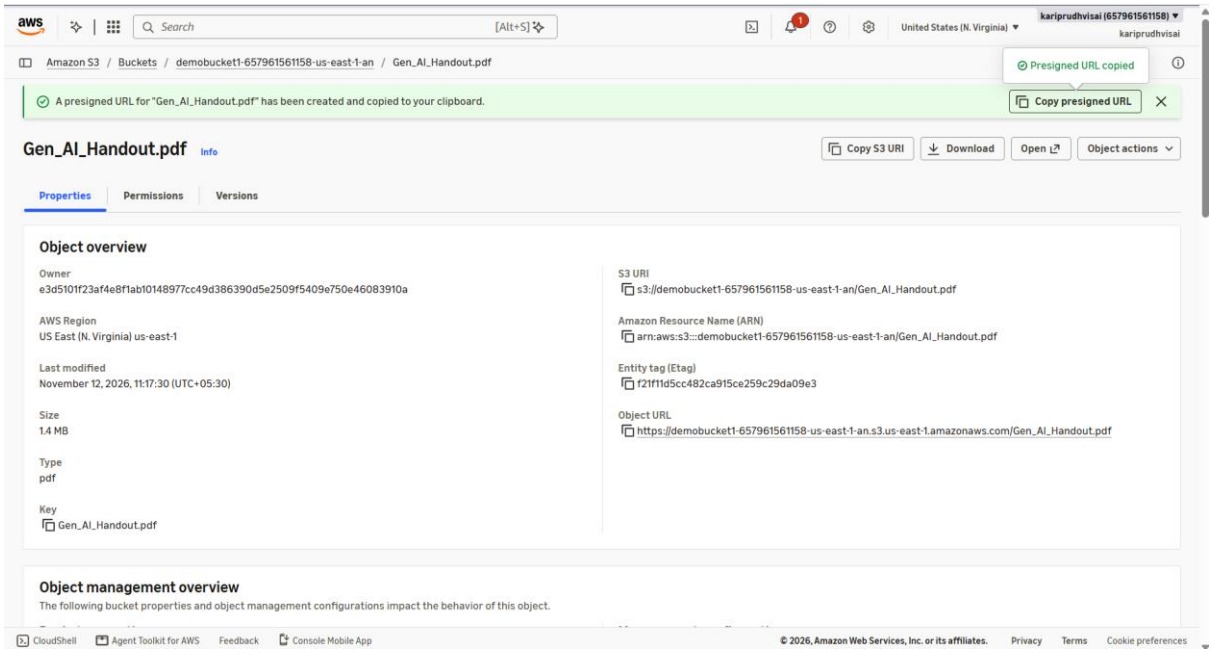
The "Create bucket" page is open again. The user has entered "amzn" into the "Bucket name" field. The "Global namespace" is selected, unlike the previous bucket creation.



A continuation of the "Create bucket" page shows the "Object Ownership" set to "ACLs disabled (recommended)". The "Block all public access" checkbox is selected.



Further down the "Create bucket" page, "Bucket Versioning" is set to "Disable," no tags are added, and "Default encryption" is set to "Server-side encryption with Amazon S3 managed keys (SSE-S3)".



The details page for the object Gen_AI_Handout.pdf within the demobucket1-657961561158-us-east-1-an bucket is shown. A success banner at the top confirms, "A presigned URL for 'Gen_AI_Handout.pdf' has been created and copied to your clipboard," and a small pop-up in the top right reiterates "Presigned URL copied".